First Moment Obstructions in Extremal, Probabilistic and Additive Combinatorics

Lecture Notes

Based on lectures by Huy Tuan Pham

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Prepared from handwritten lecture notes taken by Ying-Sien Wu.

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Abstract. If a random variable taking values in $\{0, 1, 2, \dots\}$ has expectation $o(1)$, it is zero with high probability. These notes explore how far this one-line observation can be pushed in the study of thresholds of random structures. Part I sets up thresholds, shows that the naive first moment bound can fail badly, and repairs it by taking the first moment over a cleverly chosen *cover*; the best bound obtainable in this way is the *expectation threshold*. Part II proves the Kahn–Kalai conjecture (now the Park–Pham theorem): the repaired first moment method always predicts the threshold up to a logarithmic factor. Linear programming duality turns its hypothesis into the existence of a *spread* measure, which we apply to the sunflower problem and refine to remove the logarithm. Part III leaves the complete graph: it proves robust versions of Dirac-type theorems via a spread blow-up lemma, and, in the opposite direction, constructs first moment obstructions for independent sets in sparse random Cayley graphs using additive combinatorics and Fourier analysis.

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Preface

Let X_p be a random subset of a finite set X, obtained by keeping each element independently with probability p. For a monotone property, such as "the random graph $G(N,p)$ is Hamiltonian" or "contains a K_4", the probability that X_p has the property increases from 0 to 1 as p grows, and the value of p at which it equals $\frac{1}{2}$ is the *threshold*. Locating thresholds is one of the central problems of probabilistic combinatorics. The simplest tool for the lower bound is the first moment method: if the expected number of copies of the structure is small, the structure is unlikely to appear. These notes are about a single question: *how good is the first moment method?*

The answer has three parts. The naive expectation can be misleading, because it may be inflated by rare events. It can be repaired by counting a simpler structure that every copy must contain, i.e. by taking the first moment over a *cover*. And by the Kahn–Kalai conjecture, proved by Park and Pham in 2022, the repaired method is *always* correct up to a logarithmic factor. The rest of the notes follows the consequences of this theorem in both directions: to prove that structures *do* appear one exhibits a spread measure, which certifies that no cheap cover exists; to prove that they do *not* appear one must construct the cheap cover.

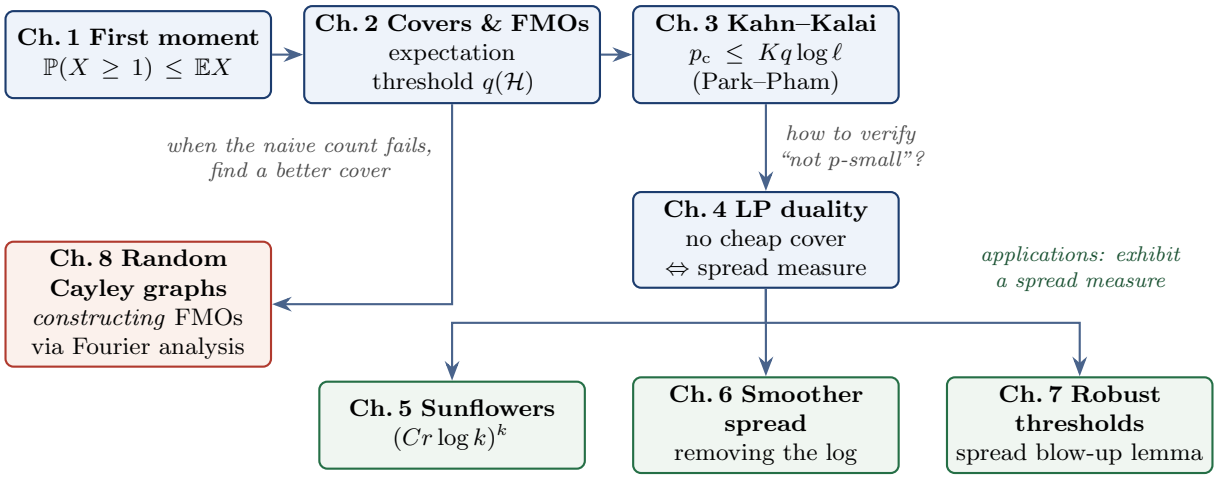
How these notes came about. The notes are based on the lecture series given by Huy Tuan Pham at the tutorial *Innovations and Challenges in Extremal Combinatorics* at the Institute for Mathematical Sciences, National University of Singapore, in July 2026. They were written up from handwritten notes taken during the lectures by Ying-Sien Wu, together with photographs of the board, and then reorganised into the present self-contained text: the narrative of the lectures is kept, arguments that were only sketched were completed, connecting remarks and figures were added, and every chapter received a section of bibliographic notes.

What the notes cover. The notes are organised in three parts.

- **Part I (The first moment method and thresholds).** The first and second moment methods, with Erdős's Ramsey bound and the independence number of $G(N,p)$ (Chapter 1); thresholds, covers, first moment obstructions and the expectation threshold, with examples showing that a logarithmic gap can occur (Chapter 2).
- **Part II (The Kahn–Kalai theorem and spread).** A complete proof of the Park–Pham theorem by the method of minimum fragments, and its sharp form due to Pham (Chapter 3); fractional covers, LP duality and spread measures (Chapter 4); the sunflower problem (Chapter 5); and Spiro's smoother spread condition, which removes the logarithm for powers of Hamilton cycles (Chapter 6).
- **Part III (Beyond the complete graph).** Robust thresholds for spanning structures in dense graphs, via regularity and a spread blow-up lemma (Chapter 7); and the independence number of sparse random Cayley graphs, where the task is to *construct* first moment obstructions, leading to the $p^{-3/2}$ bound of Alon and Pham (Chapter 8).

Chapters 1 to 3 are needed for everything that follows. Chapter 4 is used in Chapters 5 to 7, while Chapter 8 depends only on Chapters 1 and 2. The logical structure is summarised in the diagram below. Each chapter ends with a section of *Notes and references* pointing to the original papers.

Prerequisites and conventions. The prerequisites are discrete probability and elementary graph theory; the probabilistic inequalities used repeatedly are collected in Appendix A. Chapter 7 uses the regularity method, whose statements are recalled, and Chapter 8 develops the Fourier



The logical structure of the notes. Green chapters apply the Park–Pham theorem by exhibiting a spread measure; Chapter 8 goes in the opposite direction and *constructs* first moment obstructions.

analysis it needs from scratch. Throughout, $N \to \infty$ and "with high probability" (w.h.p.) means with probability $1 - o(1)$. Unless a base is indicated, log is the natural logarithm. The letters c, C, K denote absolute positive constants whose values may change from line to line. The notation is summarised on the next page.

Acknowledgments. The author thanks Huy Tuan Pham for the lectures on which these notes are based, and Ying-Sien Wu for generously sharing the handwritten notes taken during the lectures. The text was edited and integrated with the assistance of Anthropic's Claude Opus 5.5, which also helped to complete the proofs, draw the figures and cross-check the references. The notes have not been reviewed by the lecturer; as with any notes, errors may remain, and the cited papers are the authoritative sources.

Notation

We mostly follow the notation of Park and Pham [PP24a] and of Frankston, Kahn, Narayanan and Park [FKNP21], which goes back to Kahn and Kalai [KK07] and Talagrand [Tal10]. Because different papers use different symbols, the last column lists common alternatives; p_{E} and p_{f} are the symbols used in the lectures.

Symbol	Meaning	Also written
X_p	random subset of a finite set X containing each element independently with probability p	$X(p)$, μ_p
$\langle \mathcal{H} \rangle$	up-set generated by $\mathcal{H} \subseteq 2^X$: all sets containing a member of $\mathcal{H}$	–
$N_{\mathcal{H}}(S)$	number of members of $\mathcal{H}$ contained in S	–
$\mathrm{cost}_p(\mathcal{G})$	$\sum_{G \in \mathcal{G}} p^{\lvert G \rvert}$, the expected number of members of $\mathcal{G}$ in X_p	–
$\ell(\mathcal{H})$	$\max_{H \in \mathcal{H}} \lvert H \rvert$, the size of the largest member	–
$p_{\mathrm{c}}(\mathcal{H})$	threshold: the p with $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) = \frac{1}{2}$	$p_c(\mathcal{F})$
$q(\mathcal{H})$	expectation threshold: $\sup\{p : \mathcal{H} \text{ is } p\text{-small}\}$	$p_{\mathrm{E}}(\mathcal{H})$
$q_{\mathrm{f}}(\mathcal{H})$	fractional expectation threshold	$p_{\mathrm{f}}(\mathcal{H})$
$T(H, W)$	minimum fragment of H with respect to W	–
q-spread	$\mu(\{H : H \supseteq S\}) \le q^{\lvert S \rvert}$ for every set S	κ-spread, $\kappa = 1/q$

Graphs and groups

$G(N, p)$	binomial random graph on $[N] = \{1, \dots, N\}$ (often $G(n, p)$; here n is reserved for the sizes of cliques, independent sets and subsets)
G_p	random subgraph of a graph G, keeping each edge with probability p
$G(p)$	random Cayley graph on a finite abelian group G of order N, as in [AP25] (Chapter 8)
$\alpha(G)$, $\delta(G)$, $\Delta(G)$	independence number, minimum degree, maximum degree
$v(H)$, $e(H)$	number of vertices and edges of a graph H
K_r, C_N^k	complete graph on r vertices; k-th power of the N-cycle
$\binom{S}{k}$	family of k-element subsets of S
$A - A$	difference set $\{a_2 - a_1 : a_1, a_2 \in A\}$
$r(x)$, P_ℓ	number of representations $x = a_1 - a_2$; popular differences (Chapter 8)
$\widehat{f}$, $f * g$	Fourier transform and convolution on a finite abelian group

Part I

The First Moment Method and Thresholds

1 The First Moment Method

The first moment method is the most basic tool of the probabilistic method [AS16]. In this chapter we recall it together with its companion, the second moment method, and see both at work in two classical examples: Erdős's lower bound for Ramsey numbers and the independence number of $G(N,p)$. In both examples the first moment computation turns out to predict the truth exactly, which raises the question studied in the rest of the notes.

1.1 The first and second moment methods

The entire content of the first moment method is Markov's inequality.

Lemma 1.1 (First moment method). *Let X be a random variable taking values in $\mathbb{N}\cup\{0\}$. Then $\mathbb{P}(X\neq 0)\le \mathbb{E}[X]$. In particular, if $\mathbb{E}[X]=o(1)$ then $X=0$ with high probability.*

Proof. Since $X \ge \mathbf{1}\{X\ge 1\}$ pointwise, $\mathbb{P}(X\ge 1)=\mathbb{E}[\mathbf{1}\{X\ge 1\}]\le \mathbb{E}[X]$. □

The converse fails in general: $\mathbb{E}[X]\to\infty$ does not imply $X\neq 0$ w.h.p., since the expectation might be carried by a rare event on which X is huge. This possibility is the villain of these notes; we shall meet it in Example 2.6 and again, in a much more serious form, in Chapter 8. When it does *not* happen, the matching lower bound is usually supplied by the *second moment method*: by Chebyshev's inequality,

$$\mathbb{P}(X=0) \;\le\; \mathbb{P}\big(|X-\mathbb{E}X|\ge \mathbb{E}X\big) \;\le\; \frac{\operatorname{Var} X}{(\mathbb{E}X)^2} \;=\; \frac{\mathbb{E}[X^2]}{(\mathbb{E}X)^2}-1. \tag{1.1}$$

1.2 Erdős's lower bound for Ramsey numbers

The founding example of the probabilistic method is a first moment computation.

Theorem 1.2 (Erdős [Erd47]). *Almost every graph on N vertices contains neither a clique nor an independent set of size $2\log_2 N$. The same computation, with $N=\lfloor 2^{k/2}\rfloor$, shows that the Ramsey number satisfies $R(k,k)>2^{k/2}$ for all $k\ge 3$.*

Proof. A uniformly random graph on $[N]$ is $G\sim G(N,\frac12)$. Let $n=\lceil 2\log_2 N\rceil$ and, for $m\in\mathbb{N}$, let

$$N_{m\text{-clique}}=\sum_{K\in\binom{[N]}{m}}\mathbf{1}\{K \text{ is a clique in } G\},\qquad N_{m\text{-indep}}=\sum_{K\in\binom{[N]}{m}}\mathbf{1}\{K \text{ is independent in } G\}.$$

Each n-set spans a clique (or an independent set) with probability $2^{-\binom{n}{2}}$, so using $\binom{N}{n}\le (eN/n)^n$,

$$\mathbb{E}\big[N_{n\text{-clique}}+N_{n\text{-indep}}\big]=2\binom{N}{n}2^{-\binom{n}{2}}\le 2\Big(\frac{eN}{n}\,2^{-(n-1)/2}\Big)^n\le 2\Big(\frac{e\sqrt{2}}{n}\Big)^n=o(1),$$

where we used $N2^{-(n-1)/2}\le N2^{-(2\log_2 N-1)/2}=\sqrt{2}$. By Lemma 1.1, w.h.p. G has no clique and no independent set of size n. □

How good is the bound $2\log_2 N$? For $G(N,\frac12)$ it is asymptotically the truth: the second moment method (see Theorem 1.3 below with $p=\frac12$) shows that $G(N,\frac12)$ w.h.p. contains cliques and independent sets of size $(2-\varepsilon)\log_2 N$. This is the first instance of a recurring theme: *a well-chosen first moment computation predicts the truth.*

1.3 When the first moment is right: independent sets in $G(N,p)$

Theorem 1.3 ([GM75, BE76]). *Fix $p \in (0,1)$ and $\varepsilon > 0$, and put $b = 1/(1-p)$. Let $n = \lfloor (2-\varepsilon)\log_b N \rfloor$. Then w.h.p. $G(N,p)$ contains an independent set of size n. Together with the first moment bound, the independence number satisfies $\alpha(G(N,p)) = (2+o(1))\log_b N$ with high probability.*

In the lectures the statement was given in the more general form

$$n = (2-\varepsilon)\,\frac{\log(Np)}{|\log(1-p)|},$$

covering also $p = p(N) \to 0$ with $Np \to \infty$; for fixed p this is the same as the above up to an additive constant. The sparse case is true as well (Frieze [Fri90]), but the plain second moment computation below is only efficient when p is not too small; for sparse p one needs a concentration inequality in addition.

Proof. Write $q = 1-p$, so $b = 1/q$, and let $Y = N_{n\text{-indep}}(G(N,p))$. Then $\mathbb{E}Y = \binom{N}{n} q^{\binom{n}{2}}$. Since $q^{(n-1)/2} \geq q^{n/2} \geq N^{-(1-\varepsilon/2)}$,

$$\mathbb{E}Y \;\geq\; \Big(\frac{N}{n}\Big)^n q^{n(n-1)/2} \;\geq\; \Big(\frac{N^{\varepsilon/2}}{n}\Big)^n \longrightarrow \infty.$$

For the second moment, two n-sets A_1, A_2 that share k vertices are both independent with probability $q^{2\binom{n}{2}-\binom{k}{2}}$, since the $\binom{k}{2}$ pairs inside $A_1 \cap A_2$ are counted twice. Grouping pairs (A_1, A_2) by $k = |A_1 \cap A_2|$,

$$\frac{\mathbb{E}[Y^2]}{(\mathbb{E}Y)^2} = \sum_{k=0}^{n} h(k)\, q^{-\binom{k}{2}}, \qquad h(k) := \frac{\binom{n}{k}\binom{N-n}{n-k}}{\binom{N}{n}},$$

where h is the hypergeometric distribution, so $\sum_k h(k) = 1$. The terms $k \in \{0,1\}$ contribute at most $h(0) + h(1) \leq 1$. For $k \geq 2$ we use $\binom{N-n}{n-k} \leq \binom{N}{n-k}$ and

$$\frac{\binom{N}{n-k}}{\binom{N}{n}} = \prod_{i=0}^{k-1} \frac{n-i}{N-n+k-i} \leq \Big(\frac{n}{N-n}\Big)^k \leq \Big(\frac{2n}{N}\Big)^k,$$

together with $\binom{n}{k} \leq (en/k)^k$ and $q^{-\binom{k}{2}} = b^{k(k-1)/2} \leq \big(b^{(n-1)/2}\big)^k \leq (N^{1-\varepsilon/2})^k$. Hence, with $\theta := 2en^2 N^{-\varepsilon/2} \to 0$,

$$\sum_{k\geq 2} h(k)\, q^{-\binom{k}{2}} \;\leq\; \sum_{k\geq 2} \Big(\frac{2en^2 N^{1-\varepsilon/2}}{N}\Big)^k = \sum_{k\geq 2} \theta^k \leq 2\theta^2 = o(1).$$

Thus $\mathbb{E}[Y^2] \leq (1+o(1))(\mathbb{E}Y)^2$, equivalently $\mathbb{E}\big[(Y/\mathbb{E}Y - 1)^2\big] = o(1)$, and (1.1) gives $\mathbb{P}(Y = 0) = o(1)$. (In fact $Y \geq (1-\varepsilon)\mathbb{E}Y$ w.h.p. for every fixed $\varepsilon > 0$.) □

Looking ahead. In both examples the threshold sits exactly where the expected count crosses 1. Is this always so? If not, what is the correct first moment computation, and how far from the truth can it be? The rest of these notes is about these questions.

1.4 Notes and references

The probabilistic method, and the first and second moment methods in particular, are treated in the book of Alon and Spencer [AS16, Ch. 1 and 4]; the standard reference for random graphs is Janson, Łuczak and Ruciński [JŁR00]. Theorem 1.2 is from Erdős's 1947 paper [Erd47], often regarded as the beginning of the probabilistic method. The asymptotics of the clique and independence numbers of $G(N,p)$ for fixed p are due to Grimmett and McDiarmid [GM75] and Bollobás and Erdős [BE76], who proved much more precise concentration. For $p \to 0$ the plain second moment method breaks down, and the asymptotics $\alpha(G(N,p)) = (2+o(1))\log(Np)/p$ are due to Frieze [Fri90], who combined it with a martingale concentration inequality.

2 Thresholds and First Moment Obstructions

In this chapter we set up the general framework: random subsets X_p of a finite set, monotone properties and their thresholds. We then see that the naive first moment bound can be far from the truth, repair it by taking the first moment over *covers*, and arrive at the central notion of these notes, the expectation threshold. Two examples show that the expectation threshold can underestimate the threshold by a logarithmic factor.

2.1 The general setup

Let X be a finite set and $p \in [0,1]$, and recall that X_p is the random subset of X in which each element is included independently with probability p. If $X = \binom{[N]}{2}$ is the edge set of the complete graph K_N, then $X_p = G(N,p)$.

A family $\mathcal{F} \subseteq 2^X$ is *increasing* (or an *up-set*, or a *monotone property*) if $A \in \mathcal{F}$ and $B \supseteq A$ imply $B \in \mathcal{F}$. Increasing families are conveniently described by their generators: for $\mathcal{H} \subseteq 2^X$ let

$$\langle \mathcal{H} \rangle := \{S \subseteq X : \ S \supseteq H \text{ for some } H \in \mathcal{H}\}$$

be the up-set generated by $\mathcal{H}$. Every increasing $\mathcal{F}$ is of this form (take $\mathcal{H} = \mathcal{F}$, or just the inclusion-minimal members of $\mathcal{F}$), so "X_p has the monotone property $\mathcal{F}$" and "X_p contains a member of $\mathcal{H}$" are the same kind of statement.

Example 2.1. For $X = E(K_N)$:

- $\mathcal{H} = \{\text{spanning trees of } K_N\}$ gives $\langle \mathcal{H} \rangle = \{\text{connected graphs on } [N]\}$;
- $\mathcal{H} = \{\text{Hamilton cycles}\}$ gives the Hamiltonian graphs;
- $\mathcal{H} = \{\text{copies of a fixed graph } H\}$ gives the graphs containing H;
- $\mathcal{H} = \{K_{k+1}\text{-factors}\}$, i.e. vertex-disjoint copies of K_{k+1} covering $[N]$, gives the graphs containing a K_{k+1}-factor.

We call $\mathcal{H}$ *trivial* if $\langle \mathcal{H} \rangle = \emptyset$ (that is, $\mathcal{H} = \emptyset$) or $\langle \mathcal{H} \rangle = 2^X$ (that is, $\emptyset \in \mathcal{H}$), and we tacitly assume nontriviality whenever it matters.

2.2 Thresholds

For nontrivial $\mathcal{H}$ the function $f(p) := \mathbb{P}(X_p \in \langle \mathcal{H} \rangle)$ is a polynomial in p with $f(0) = 0$ and $f(1) = 1$, and it is strictly increasing on $[0,1]$ (for instance by Russo's formula, or by the coupling below). This justifies the following definition.

Definition 2.2 (Threshold). The *threshold* $p_c(\mathcal{H})$ of a nontrivial $\mathcal{H} \subseteq 2^X$ is the unique $p \in (0,1)$ with $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) = \frac{1}{2}$.

The simplest structural fact about thresholds uses an idea that will be the engine of the proof of the Kahn–Kalai conjecture: *a random set of density about mp can be generated as a union of m independent random sets of density p.*

Lemma 2.3 (Independent rounds). *Let $W_1, \dots, W_m$ be independent copies of X_p. Then $W_1 \cup \dots \cup W_m$ has the distribution of $X_{p'}$ with $p' = 1 - (1-p)^m \le mp$. Consequently, for every increasing family $\mathcal{F}$,*

$$1 - \mathbb{P}\big(X_{p'} \in \mathcal{F}\big) \ \le \ \big(1 - \mathbb{P}(X_p \in \mathcal{F})\big)^m.$$

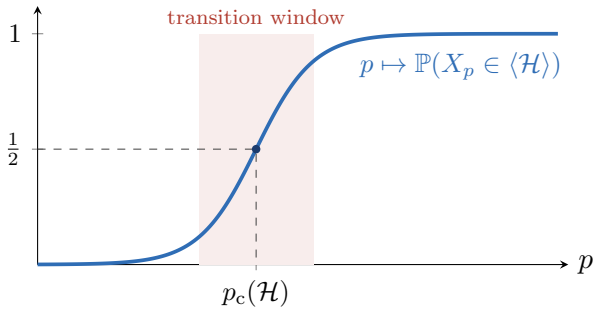


Figure 2.1: The probability that X_p has a monotone property increases from 0 to 1; the threshold p_c is where it equals $\frac{1}{2}$. By Theorem 2.4, the transition always happens within a constant factor of p_c.

Proof. Each element lies outside all m sets independently with probability $(1-p)^m$. If some $W_i \in \mathcal{F}$ then the union is in $\mathcal{F}$, so the union fails only if every W_i fails. □

Theorem 2.4 (Bollobás–Thomason [BT87]). *For every nontrivial $\mathcal{H}$ and every integer $m \geq 1$,*

$$\mathbb{P}\big(X_{p_c/m} \in \langle \mathcal{H} \rangle\big) \leq 1 - 2^{-1/m} \leq \frac{\log 2}{m} \qquad \text{and} \qquad \mathbb{P}\big(X_{\min(1,mp_c)} \in \langle \mathcal{H} \rangle\big) \geq 1 - 2^{-m}.$$

In particular, for any sequence of nontrivial monotone properties, $p \ll p_c$ implies $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \to 0$ and $p \gg p_c$ implies $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \to 1$.

Proof. Write $f(p) = \mathbb{P}(X_p \in \langle \mathcal{H} \rangle)$, which is nondecreasing. Applying Lemma 2.3 with $p = p_c$ and using $1-(1-p_c)^m \leq mp_c$ gives $1 - f(mp_c) \leq (1-f(p_c))^m = 2^{-m}$. Applying it with $p = p_c/m$ and using $1-(1-p_c/m)^m \leq p_c$ gives $\frac{1}{2} = 1 - f(p_c) \leq (1-f(p_c/m))^m$, which rearranges to the first inequality. □

So every monotone property has a threshold in the coarse sense of Figure 2.1, and the problem is to *locate* $p_c(\mathcal{H})$, at least up to constant or logarithmic factors. (For many natural properties the transition window is in fact much narrower than p_c; this is the theory of *sharp thresholds* [FK96], which we will not need.)

2.3 The naive first moment bound

For $S \subseteq X$ let $N_{\mathcal{H}}(S) := \#\{H \in \mathcal{H} : H \subseteq S\}$ be the number of members of $\mathcal{H}$ in S. Since $X_p \in \langle \mathcal{H} \rangle$ exactly when $N_{\mathcal{H}}(X_p) \geq 1$, Lemma 1.1 gives

$$\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \;\leq\; \mathbb{E}\big[N_{\mathcal{H}}(X_p)\big] \;=\; \sum_{H \in \mathcal{H}} p^{|H|} \;=\; \mathrm{cost}_p(\mathcal{H}). \tag{2.1}$$

Hence $\mathrm{cost}_p(\mathcal{H}) < \frac{1}{2}$ implies $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) < \frac{1}{2}$, that is, $p_c(\mathcal{H}) > p$.

Example 2.5 *(Fixed subgraphs).* Let H be a fixed graph with $v(H)$ vertices and $e(H)$ edges and let $\mathcal{H}$ be the family of copies of H in K_N. There are $\Theta(N^{v(H)})$ copies, so $\mathrm{cost}_p(\mathcal{H}) = \Theta(N^{v(H)}p^{e(H)})$ and therefore

$$p_c(\mathcal{H}) \;\geq\; \Omega\big(N^{-v(H)/e(H)}\big).$$

For $H = K_4$ this gives $p_c \geq \Omega(N^{-2/3})$, and the second moment method shows that this is sharp: $p_c(\{\text{copies of } K_4\}) = \Theta(N^{-2/3})$.

Example 2.6 *(The naive bound can be too weak).* Let K_4^+ be K_4 with a pendant edge attached (Figure 2.2), so $v = 5$ and $e = 7$. Example 2.5 gives only $p_c \geq \Omega(N^{-5/7})$. But every graph

containing K_4^+ contains K_4, so in fact $p_c(K_4^+) \geq p_c(K_4) = \Omega(N^{-2/3})$, which is far larger since $5/7 > 2/3$.

What goes wrong in the window $N^{-5/7} \ll p \ll N^{-2/3}$? There $\mathbb{E}[\#K_4] = \Theta(N^4 p^6) \to 0$, while $\mathbb{E}[\#K_4^+] = \Theta(N^5 p^7) = \Theta(N^4 p^6 \cdot Np) \to \infty$. A copy of K_4 is a *rare* event, but when it occurs it typically comes with about $4Np \to \infty$ pendant edges, each producing a copy of K_4^+. The expectation of $\#K_4^+$ is inflated by a rare event on which the count is huge.

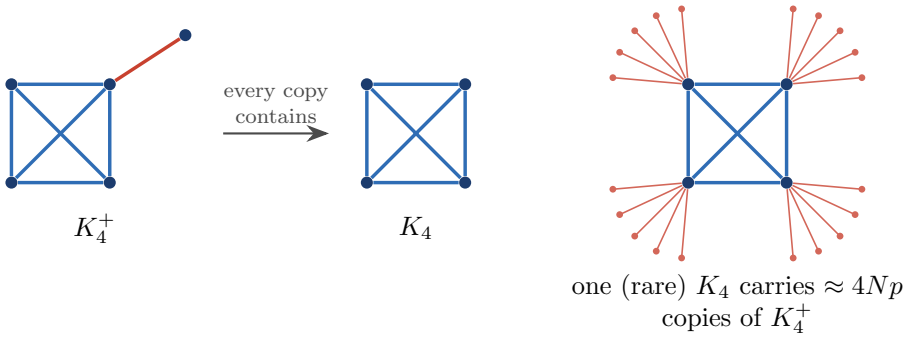


Figure 2.2: Left: the copies of K_4 form a *cover* of the copies of K_4^+. Right: why the naive count for K_4^+ is misleading when $N^{-5/7} \ll p \ll N^{-2/3}$.

2.4 Covers and first moment obstructions

Example 2.6 suggests the remedy: do the first moment computation not for $\mathcal{H}$ itself, but for a family of smaller sets that every member of $\mathcal{H}$ must contain.

Definition 2.7 (Cover). Let $\mathcal{H}, \mathcal{G} \subseteq 2^X$. We say that $\mathcal{G}$ *covers* $\mathcal{H}$ if every $H \in \mathcal{H}$ contains some $G \in \mathcal{G}$; equivalently $\mathcal{H} \subseteq \langle \mathcal{G} \rangle$, or $\langle \mathcal{H} \rangle \subseteq \langle \mathcal{G} \rangle$.

If $\mathcal{G}$ covers $\mathcal{H}$, then by (2.1) applied to $\mathcal{G}$,

$$\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \;\leq\; \mathbb{P}(X_p \in \langle \mathcal{G} \rangle) \;\leq\; \mathrm{cost}_p(\mathcal{G}). \tag{2.2}$$

For instance, the copies of K_4 cover the copies of K_4^+, which recovers $p_c(K_4^+) = \Omega(N^{-2/3})$.

Definition 2.8 (First moment obstruction; p-small). Let $\langle \mathcal{H} \rangle$ be the target property. A cover $\mathcal{G}$ of $\mathcal{H}$ with $\mathrm{cost}_p(\mathcal{G}) < \frac{1}{2}$ is called a *first moment obstruction* (FMO) at p. We say that $\mathcal{H}$ is *p-small* if there is a first moment obstruction at p.

Proposition 2.9. *If $\mathcal{H}$ is p-small, then $p_c(\mathcal{H}) > p$.*

Proof. By (2.2), $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \leq \mathrm{cost}_p(\mathcal{G}) < \frac{1}{2}$. □

Two trivial remarks will be used repeatedly: $\mathcal{H}$ always covers itself, so the naive bound is the special case $\mathcal{G} = \mathcal{H}$; and if $\mathcal{H}$ is p-small then it is p'-small for all $p' < p$, since costs are increasing in p.

Example 2.10 *(Fixed subgraphs, revisited).* Let $\mathcal{H}$ be the copies of H in K_N. For every subgraph $K \subseteq H$ with $e(K) > 0$, the copies of K cover $\mathcal{H}$, and they form an FMO at $p = cN^{-v(K)/e(K)}$. Choosing the densest K gives $p_c(\mathcal{H}) \geq \Omega(N^{-1/m(H)})$, where $m(H) := \max\{e(K)/v(K) : K \subseteq H,\ v(K) > 0\}$. Bollobás [Bol81] proved that this is the truth:

$$p_c(\{\text{copies of } H\}) = \Theta\big(N^{-1/m(H)}\big).$$

So for small subgraphs, first moment obstructions coming from subgraphs tell the whole story, and the matching upper bound comes from the second moment method.

2.5 The expectation threshold

Definition 2.11 (Expectation threshold). The *expectation threshold* of $\mathcal{H}$ is

$$q(\mathcal{H}) := \sup\{p : \ \mathcal{H} \text{ is } p\text{-small}\}.$$

By Proposition 2.9, $q(\mathcal{H}) \le p_c(\mathcal{H})$ always: $q(\mathcal{H})$ is the best lower bound on the threshold that can be obtained from the first moment method applied to covers. Can the gap between them be large? The next two examples show that a logarithmic gap can occur.

Example 2.12 *(Coupon collector).* Let $X = B_1 \sqcup \cdots \sqcup B_k$ be a disjoint union of k blocks of size m, and let $\mathcal{H}$ be the family of *transversals*, i.e. sets meeting every block in exactly one element (Figure 2.3). Then $X_p \in \langle \mathcal{H} \rangle$ if and only if X_p meets every block, so

$$\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) = \big(1 - (1-p)^m\big)^k.$$

For $p = a \log k/m$ we have $(1-p)^m \approx k^{-a}$, so this tends to 0 if $a < 1$ and to 1 if $a > 1$ (as $k \to \infty$ with $m \gg \log k$). Hence $p_c(\mathcal{H}) \sim \log k/m$. On the other hand, the singletons of one block cover $\mathcal{H}$ and cost mp, so $\mathcal{H}$ is p-small for $p < 1/(2m)$; and we shall see in Example 4.4 that $\mathcal{H}$ is *not* $(1/m)$-small. Thus $q(\mathcal{H}) = \Theta(1/m)$ and

$$\frac{p_c(\mathcal{H})}{q(\mathcal{H})} = \Theta(\log k), \qquad k = \max_{H \in \mathcal{H}} |H|.$$

The obstruction at $p \approx \log k/m$, namely an empty block, is a coupon collector phenomenon, which no first moment over covers can see.

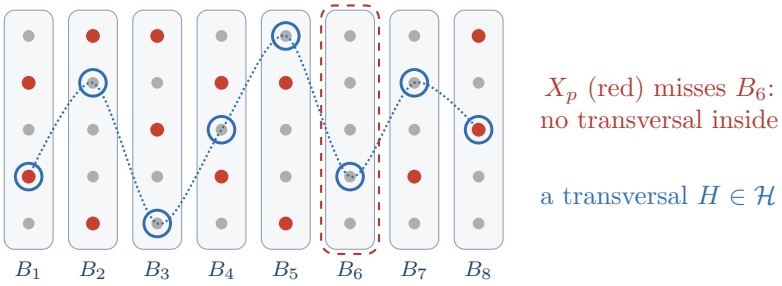


Figure 2.3: The coupon collector example: k blocks of size m. The expectation threshold is $\Theta(1/m)$ but the threshold is $\sim \log k/m$, because at density $p \ll \log k/m$ some block is empty.

Example 2.13 *(Hamilton paths).* Let $X = E(K_N)$ and $\mathcal{H} = \{\text{Hamilton paths}\}$. The naive bound gives $\text{cost}_p(\mathcal{H}) = \frac{N!}{2} p^{N-1}$, which is tiny for $p = c/N$ with $c < e$. A simpler FMO: every Hamilton path passes through vertex 1, so the star $\mathcal{G} = \big\{\{1, v\} : v \ne 1\big\}$ (as a family of singletons) covers $\mathcal{H}$, at cost $(N-1)p$. Either way $q(\mathcal{H}) \ge c/N$, and conversely we shall see in Example 4.5 that $\mathcal{H}$ is not (C/N)-small for a suitable constant C, so $q(\mathcal{H}) = \Theta(1/N)$.

However, for $p = (1-\varepsilon)\log N/N$ the graph $G(N,p)$ has isolated vertices w.h.p. [ERe60], and so contains no Hamilton path. Hence $p_c(\mathcal{H}) \ge (1 - o(1)) \log N/N$, although $\mathcal{H}$ is not C/N-small. (In fact $p_c(\mathcal{H}) \sim \log N/N$ [Pós76, KS83].) Again the obstruction, an isolated vertex, is of coupon collector type, and the gap $\log N$ equals $\log \ell$ up to a constant factor, where $\ell = N - 1$ is the size of the members of $\mathcal{H}$.

Looking ahead. The examples show that $p_c(\mathcal{H})/q(\mathcal{H})$ can be as large as $\log \ell$, where ℓ is the size of the largest member of $\mathcal{H}$. Kahn and Kalai conjectured that this is the worst that can happen: *the first moment method over covers always predicts the threshold up to a logarithmic factor.* This is the subject of the next chapter.

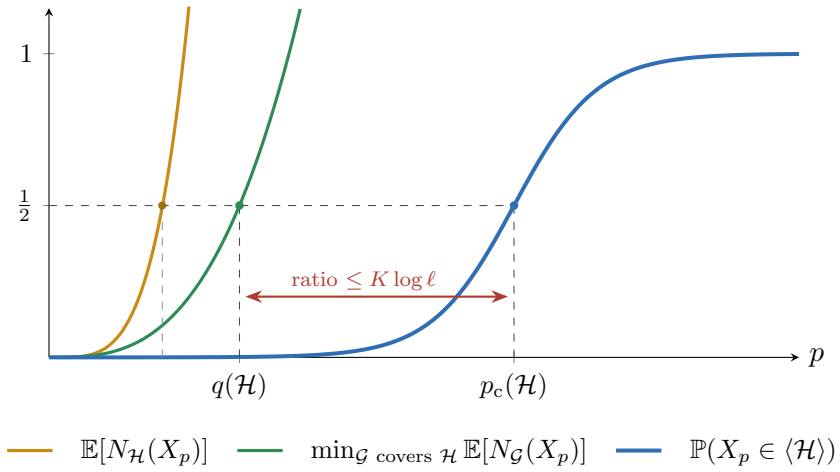


Figure 2.4: Three curves, ordered by (2.2): the naive expectation (gold), the cheapest cover (green), and the true probability (blue). Each crosses $\frac{1}{2}$ to the right of the previous one. The green crossing is the expectation threshold $q(\mathcal{H})$, the blue one is $p_c(\mathcal{H})$; by Theorem 3.1 they differ by at most a factor $K \log \ell$.

2.6 Notes and references

Thresholds of random graph properties go back to Erdős and Rényi [ERe60], who also determined the threshold for balanced fixed subgraphs; Bollobás [Bol81] settled the general case of Example 2.10. That every monotone property has a threshold is due to Bollobás and Thomason [BT87]; the argument via independent rounds (Lemma 2.3) is the standard proof, and it reappears at the heart of Chapter 3. For sharp thresholds see Friedgut and Kalai [FK96]. Covers and the expectation threshold $q(\mathcal{H})$ are from Kahn and Kalai [KK07]; the term *p-small* is Talagrand's [Tal10], and the notation $q(\mathcal{H})$ follows [KK07, FKNP21, PP24a]. (The symbol p_{E}, used for the expectation threshold in the lectures, denotes in [KK07] a different quantity, defined only through expected counts of subgraphs; it is the subject of the so-called second Kahn–Kalai conjecture.) The term *first moment obstruction* follows the lectures. The threshold for Hamiltonicity, of which Example 2.13 is the path version, is due to Pósa [Pós76] and Komlós and Szemerédi [KS83].

Part II

The Kahn–Kalai Theorem and Spread

3 The Kahn–Kalai Conjecture

We now state and prove the Kahn–Kalai conjecture. The proof of Park and Pham is short but subtle: after explaining the idea, we introduce minimum fragments, prove the key lemma by an encoding argument, and iterate it about $\log_2 \ell$ times. At the end of the chapter we see that the same argument, stopped early, already shows that a family which is not p-small is "almost hit" at density $O(sp)$, and we state Pham's sharp form of this phenomenon.

3.1 Statement

Theorem 3.1 (Kahn–Kalai conjecture [KK07]; Park–Pham [PP24a]). *Let $\mathcal{H} \subseteq 2^X$ and let $\ell \geq 1$ be an integer such that $|H| \leq \ell$ for all $H \in \mathcal{H}$. If $\mathcal{H}$ is not p-small, then*

$$\mathbb{P}\big(X_{p'} \in \langle \mathcal{H} \rangle\big) \;\geq\; \frac{1}{2}, \qquad p' = \min\{1,\; 100\, p \log_2(2\ell)\}.$$

Consequently, writing $\ell(\mathcal{H}) = \max_{H \in \mathcal{H}} |H|$, every nontrivial $\mathcal{H}$ satisfies

$$q(\mathcal{H}) \;\leq\; p_c(\mathcal{H}) \;\leq\; 100\, q(\mathcal{H}) \log_2\big(2\ell(\mathcal{H})\big).$$

The second statement follows from the first: for every $p > q(\mathcal{H})$ the family is not p-small, so $p_c(\mathcal{H}) \leq 100\, p \log_2(2\ell)$; now let $p \downarrow q(\mathcal{H})$. (The constant 100 is what our proof gives and has not been optimised. The lectures stated the bound with $\log |X|$; since $\ell \leq |X|$, the bound with $\log \ell$ is stronger.)

Remark 3.2 *(On the statement).*

(a) *The logarithm is necessary*, by the coupon collector (Example 2.12) and Hamilton path (Example 2.13) examples.

(b) *Why it is useful.* The threshold p_c is a "dynamic" quantity: it concerns a random set and requires understanding correlations. The expectation threshold is "static": it is a purely combinatorial covering problem. Theorem 3.1 says that up to a logarithm the dynamic question *reduces* to the static one. Combined with the tools of Chapter 4, it gives in a few lines the thresholds, up to the log factor, for Hamiltonicity, bounded-degree spanning trees, K_r-factors, hypergraph perfect matchings, and many other spanning structures, questions that previously each required a dedicated and often difficult argument.

3.2 The idea of the proof

Let $s \approx \log_2 \ell$ and let $W_1, \dots, W_s$ be independent copies of X_{cp} for a large constant c. By Lemma 2.3, $W_1 \cup \cdots \cup W_s$ is distributed like $X_{p'}$ with $p' \leq csp$, which is the density in Theorem 3.1. We would like $W_1 \cup \cdots \cup W_s \in \langle \mathcal{H} \rangle$; call the outcome *bad* if $W_1 \cup \cdots \cup W_s \notin \langle \mathcal{H} \rangle$. We shall exploit the tension between

$$W_1 \cup \cdots \cup W_s \notin \langle \mathcal{H} \rangle \qquad \text{and} \qquad W_1 \cup \cdots \cup W_s \cup H \in \langle \mathcal{H} \rangle \;\text{ for every } H \in \mathcal{H}$$

to *construct from the random sets W_i a small cover of $\mathcal{H}$*: a random family $\mathcal{U}$ that covers $\mathcal{H}$ whenever the outcome is bad, and whose expected cost is tiny. If bad outcomes had probability at least $\frac{1}{2}$, then with positive probability $\mathcal{U}$ would be a cover of cost less than $\frac{1}{2}$, that is, a first moment obstruction, contradicting the assumption that $\mathcal{H}$ is not p-small.

The philosophy. The proof is a contrapositive: *from the failure of $X_{Cp\log\ell}$ to contain a member of $\mathcal{H}$, we extract a first moment obstruction at p.* The first moment obstruction is built out of the random set itself.

How does W_1 help? After exposing W_1, a member $H \in \mathcal{H}$ "only needs" $H \setminus W_1$ to be completed. But one can do better: we may switch to a different member $H' \in \mathcal{H}$ as long as H' is available inside $W_1 \cup H$. This leads to the notion of minimum fragments.

3.3 Minimum fragments

Definition 3.3 (Feasible sets and minimum fragments). A set $Z \subseteq X$ is *feasible* if $Z \in \langle \mathcal{H} \rangle$. For $H \in \mathcal{H}$ and $W \subseteq X$, the *minimum fragment* $T(H, W)$ is a minimum-size subset $T \subseteq H$ such that $W \cup T$ is feasible. (Such T exists since $T = H$ works; ties are broken by a fixed rule, so that $T(H, W)$ is a function of (H, W).)

The following properties are immediate from minimality:

- $T(H, W) \cap W = \emptyset$, and $T(H, W) = H' \setminus W$ for some $H' \in \mathcal{H}$ with $H' \subseteq W \cup H$;
- $T(H, W) = \emptyset$ if and only if W is feasible.

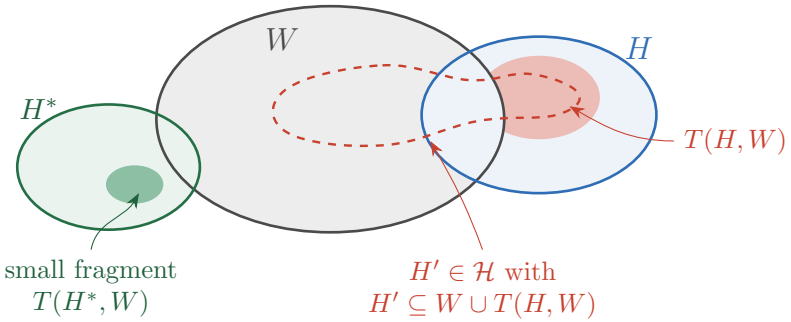


Figure 3.1: Minimum fragments. The red region $T(H, W) \subseteq H \setminus W$ is a smallest piece of H which, added to W, completes *some* member H' of $\mathcal{H}$ (dashed). Some members, like H^*, have only a small fragment left after W is exposed; others have a large one.

The plan is now: after exposing W_1, split the members of $\mathcal{H}$ according to whether their minimum fragment is *large* ($|T| > \ell/2$) or *small* ($|T| \le \ell/2$).

- Large fragments are *cheap in aggregate*: the family of large fragments, which covers the corresponding members of $\mathcal{H}$, has small expected cost. This is the key lemma below.
- Small fragments form a new family of sets of size at most $\ell/2$. Any W_2 containing a small fragment completes, together with W_1, a member of $\mathcal{H}$. So we can iterate, halving ℓ in each round.

3.4 The key lemma

Lemma 3.4 (Key lemma). *Let $\ell > 0$ be real and let $\mathcal{H} \subseteq 2^X$ consist of sets of size at most ℓ. Let $c \ge 1$ and $p \in (0, 1)$ with $cp \le 1$, let $W \sim X_{cp}$, and let*

$$\mathcal{U}_W := \big\{T(H, W) : \ H \in \mathcal{H},\ |T(H, W)| > \ell/2\big\}$$

be the family of large minimum fragments. Then

$$\mathbb{E}_{W \sim X_{cp}}\big[\mathrm{cost}_p(\mathcal{U}_W)\big] = \mathbb{E}_{W \sim X_{cp}}\Big[\sum_{T \in \mathcal{U}_W} p^{|T|}\Big] \ \le \ \Big(\frac{4}{c}\Big)^{\ell/2}.$$

The proof is an *encoding argument* (see Figure 3.2). For every feasible Z, fix once and for all a member $\widehat{H}(Z) \in \mathcal{H}$ with $\widehat{H}(Z) \subseteq Z$.

Claim 3.5. Let $H \in \mathcal{H}$, $W \subseteq X$, $T = T(H, W)$ and $Z = W \cup T$. Then $T \subseteq \widehat{H}(Z)$.

Proof. Since $\widehat{H}(Z) \subseteq Z = W \cup T$, every element of $\widehat{H}(Z)$ lies in W or in $T \cap \widehat{H}(Z)$, so $\widehat{H}(Z) \subseteq W \cup (T \cap \widehat{H}(Z))$. Thus $W \cup (T \cap \widehat{H}(Z))$ is feasible, and $T \cap \widehat{H}(Z) \subseteq T \subseteq H$. By the minimality of $|T|$ we get $T \cap \widehat{H}(Z) = T$, that is, $T \subseteq \widehat{H}(Z)$. □

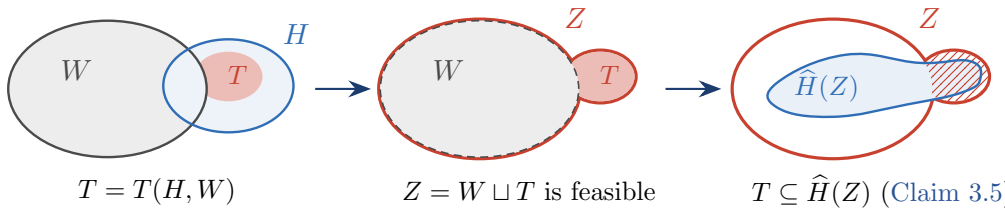


Figure 3.2: The encoding behind Lemma 3.4. The pair (W, T) is recovered from $Z = W \sqcup T$ together with the position of T inside the fixed set $\widehat{H}(Z)$, which has size at most ℓ.

Proof of Lemma 3.4. Write $P(W) := \mathbb{P}(X_{cp} = W) = (cp)^{|W|}(1-cp)^{|X|-|W|}$. Grouping the large fragments by size,

$$\mathbb{E}\big[\mathrm{cost}_p(\mathcal{U}_W)\big] = \sum_{W \subseteq X} P(W) \sum_{T \in \mathcal{U}_W} p^{|T|} = \sum_{t > \ell/2} p^t \sum_{(W,T)} P(W),$$

where the inner sum runs over pairs (W, T) such that $T = T(H, W)$ for some $H \in \mathcal{H}$ and $|T| = t$. Map such a pair to (Z, T) with $Z = W \sqcup T$. This map is injective (since $W = Z \setminus T$), Z is feasible, and $T \subseteq \widehat{H}(Z)$ with $|T| = t$ by Claim 3.5. Moreover, since $|W| = |Z| - t$,

$$P(W) = P(Z)\Big(\frac{1-cp}{cp}\Big)^t \le P(Z)\,(cp)^{-t}.$$

Since $|\widehat{H}(Z)| \le \ell$, a feasible Z is the image of at most $\binom{\lfloor \ell \rfloor}{t}$ pairs with $|T| = t$. Therefore

$$\begin{aligned}\mathbb{E}\big[\mathrm{cost}_p(\mathcal{U}_W)\big] &\le \sum_{t > \ell/2} p^t (cp)^{-t} \binom{\lfloor \ell \rfloor}{t} \sum_{Z \text{ feasible}} P(Z) \\ &\le \sum_{t > \ell/2} \binom{\lfloor \ell \rfloor}{t} c^{-t} \;\le\; c^{-\ell/2}\, 2^{\ell} = \Big(\frac{4}{c}\Big)^{\ell/2}.\end{aligned}$$

□

Remark 3.6 *(Why this works).* Removing the t elements of T from Z multiplies the probability by about $(cp)^{-t}$; this loss is paid for by the weight p^t of T in the cost, up to a factor c^{-t}. Locating T inside $\widehat{H}(Z)$ costs at most 2^{ℓ}. Because the fragment is *large*, $c^{-t} \le c^{-\ell/2}$ beats 2^{ℓ} once $c > 4$. Note that the bound does not depend on $|X|$ or $|\mathcal{H}|$ at all. Small fragments cannot be treated this way, so they are passed on to the next round.

3.5 Iteration: proof of Theorem 3.1

Proof of Theorem 3.1. Let $c = 100$. If $p' = 1$ there is nothing to prove ($\mathcal{H} \neq \emptyset$ since $\mathcal{H}$ is not p-small, so $X \in \langle \mathcal{H} \rangle$); so assume $100p \log_2(2\ell) < 1$, which gives $cp < 1$. Let $s = \lfloor \log_2 \ell \rfloor + 1$, so that $2^{s-1} \le \ell < 2^s$ and $s \le \log_2(2\ell)$, and let $W_1, \dots, W_s$ be independent copies of X_{cp}. Put $\ell_i := \ell / 2^i$.

Construction. Set $\mathcal{H}_0 := \mathcal{H}$. For $i = 1, \dots, s$, given $\mathcal{H}_{i-1}$ (a family of sets of size at most ℓ_{i-1}, determined by $W_1, \dots, W_{i-1}$), take minimum fragments *with respect to* $\mathcal{H}_{i-1}$ and W_i: for $H \in \mathcal{H}_{i-1}$ let $T_i(H)$ be a minimum-size subset $T \subseteq H$ with $W_i \cup T \in \langle \mathcal{H}_{i-1} \rangle$, and set

$$\mathcal{U}_i := \{T_i(H) : \ H \in \mathcal{H}_{i-1},\ |T_i(H)| > \ell_{i-1}/2\} \quad \text{(large fragments)},$$
$$\mathcal{H}_i := \{T_i(H) : \ H \in \mathcal{H}_{i-1},\ |T_i(H)| \le \ell_{i-1}/2\} \quad \text{(small fragments)}.$$

Then $\mathcal{H}_i$ consists of sets of size at most ℓ_i. Finally let $\mathcal{U} := \mathcal{U}_1 \cup \cdots \cup \mathcal{U}_s$ and $W := W_1 \cup \cdots \cup W_s$.

Two invariants. By induction on i:

(i) every $H \in \mathcal{H}$ contains a member of $\mathcal{U}_1 \cup \cdots \cup \mathcal{U}_i \cup \mathcal{H}_i$;

(ii) if $Y \in \langle \mathcal{H}_i \rangle$, then $W_1 \cup \cdots \cup W_i \cup Y \in \langle \mathcal{H} \rangle$.

For (i): if H contains $H'' \in \mathcal{H}_{i-1}$, then $T_i(H'') \subseteq H'' \subseteq H$ and $T_i(H'') \in \mathcal{U}_i \cup \mathcal{H}_i$. For (ii): if $Y \supseteq T$ with $T = T_i(H'') \in \mathcal{H}_i$, then $W_i \cup Y \supseteq W_i \cup T \in \langle \mathcal{H}_{i-1} \rangle$, and we apply the induction hypothesis.

The end. Since $\ell_s < 1$, we have $\mathcal{H}_s \subseteq \{\emptyset\}$. If $\mathcal{H}_s = \{\emptyset\}$ then (ii) with $Y = \emptyset$ gives $W \in \langle \mathcal{H} \rangle$. If $\mathcal{H}_s = \emptyset$ then (i) says that $\mathcal{U}$ covers $\mathcal{H}$, and since $\mathcal{H}$ is not p-small, $\mathrm{cost}_p(\mathcal{U}) \ge \frac{1}{2}$. Hence

$$\mathbb{P}(W \notin \langle \mathcal{H} \rangle) \ \le \ \mathbb{P}\big(\mathrm{cost}_p(\mathcal{U}) \ge \tfrac{1}{2}\big) \ \le \ 2\,\mathbb{E}[\mathrm{cost}_p(\mathcal{U})]. \tag{3.1}$$

The cost. Conditionally on $W_1, \dots, W_{i-1}$, the family $\mathcal{H}_{i-1}$ is fixed and $W_i \sim X_{cp}$ is independent of it, so Lemma 3.4 (with ℓ_{i-1} in place of ℓ) gives $\mathbb{E}[\mathrm{cost}_p(\mathcal{U}_i)] \le (4/c)^{\ell_{i-1}/2} = x^{\ell/2^{i-1}}$ with $x := (4/c)^{1/2} = \frac{1}{5}$. Since $\ell/2^{i-1} \ge 2^{s-i}$, putting $j = s - i$,

$$\mathbb{E}[\mathrm{cost}_p(\mathcal{U})] \le \sum_{i=1}^{s} \mathbb{E}[\mathrm{cost}_p(\mathcal{U}_i)] \le \sum_{j=0}^{s-1} x^{2^j} \le \sum_{m \ge 1} x^m = \frac{x}{1-x} = \frac{1}{4}.$$

By (3.1), $\mathbb{P}(W \in \langle \mathcal{H} \rangle) \ge \frac{1}{2}$. Finally, by Lemma 2.3, W is distributed as $X_{p''}$ with $p'' \le csp \le 100p \log_2(2\ell) = p'$, and since $\langle \mathcal{H} \rangle$ is increasing, $\mathbb{P}(X_{p'} \in \langle \mathcal{H} \rangle) \ge \mathbb{P}(X_{p''} \in \langle \mathcal{H} \rangle) \ge \frac{1}{2}$. □

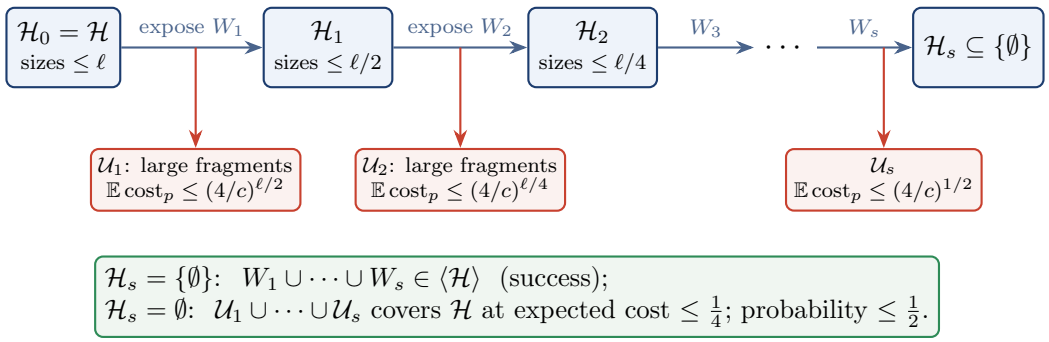


Figure 3.3: The iteration in the proof of Theorem 3.1. In each round the members whose minimum fragment is large are covered by a cheap family $\mathcal{U}_i$; the others are replaced by their fragments, of half the size. After $s \approx \log_2 \ell$ rounds nothing is left.

3.6 Partial covering and a sharp form

Stopping the iteration early already gives more information. If we run only s rounds, for any $1 \le s \le \lfloor \log_2 \ell \rfloor + 1$, the same bound $\mathbb{E}[\mathrm{cost}_p(\mathcal{U})] \le \frac{1}{4}$ holds. On the event that $\mathcal{U}$ does not cover $\mathcal{H}$, which has probability at least $\frac{1}{2}$, some member of $\mathcal{H}$ contains a set $T \in \mathcal{H}_s$, and then by invariant (ii) some $H' \in \mathcal{H}$ satisfies $H' \subseteq W \cup T$ with $|T| \le \ell/2^s$. We obtain:

Corollary 3.7. *Let $\mathcal{H}$ consist of sets of size at most ℓ, and suppose $\mathcal{H}$ is not p-small. For every integer $s \geq 1$, with probability at least $\frac{1}{2}$ the set $W \sim X_{\min(1,100sp)}$ satisfies $|H \setminus W| \leq 2^{-s}\ell$ for some $H \in \mathcal{H}$. In particular, if $\mathcal{H}$ is ℓ-uniform, then*

$$\mathbb{P}\Big(\max_{H\in\mathcal{H}} \frac{|W \cap H|}{|H|} \;\geq\; 1 - 2^{-s}\Big) \;\geq\; \frac{1}{2}.$$

(For $s > \lfloor \log_2 \ell \rfloor + 1$ the statement follows from Theorem 3.1.) So a family that is not p-small is "almost hit" at density $O(sp)$, with an exponentially small deficiency in s. Pham showed that this holds for arbitrary families and, more generally, for arbitrary weightings of the members of $\mathcal{H}$.

Theorem 3.8 (Pham [Pha25]). *There is an absolute constant C such that the following holds. Suppose $\mathcal{H} \subseteq 2^X$ is not p-small, and for each $H \in \mathcal{H}$ let $\lambda_H : H \to \mathbb{R}_{\geq 0}$ satisfy $\sum_{x\in H} \lambda_H(x) = 1$. Then for every integer $s \geq 1$, with probability at least $\frac{1}{2}$,*

$$\max_{H\in\mathcal{H}} \sum_{x\in H\cap X_{Csp}} \lambda_H(x) \;\geq\; 1 - 2^{-s}.$$

In [Pha25, Theorem 1.3] the conclusion is stated with probability $\frac{1}{3}$ for X_{16sp}. Since the event is increasing, two independent rounds (Lemma 2.3) raise the probability to $1 - (\frac{2}{3})^2 > \frac{1}{2}$ at density $32sp$.

The uniform weights $\lambda_H \equiv 1/|H|$ give, for an arbitrary (not necessarily uniform) $\mathcal{H}$ that is not p-small: with probability at least $\frac{1}{2}$, $\max_H |X_{Csp} \cap H|/|H| \geq 1 - 2^{-s}$. Taking $2^s > \ell$ forces $|H \setminus X_{Csp}| < 1$, so Theorem 3.8 contains Theorem 3.1. The weighted statement is a sharp form of Talagrand's *selector process conjecture*, which Park and Pham proved in [PP24b] with an absolute constant in place of $1 - 2^{-s}$; the proof of Theorem 3.8 refines the minimum fragment method with "towers" of minimum fragments.

Remark 3.9 *(The rate 2^{-s} is optimal).* Consider the transversals of Example 2.12 with $p = 1/m$; we shall see in Example 4.4 that this $\mathcal{H}$ is not p-small. For $p' = Csp \leq \frac{1}{2}$ each block is missed by $X_{p'}$ independently with probability $(1-p')^m \geq e^{-2p'm} = e^{-2Cs}$. The best transversal meets exactly the blocks that $X_{p'}$ meets, so as $k \to \infty$,

$$\max_{H\in\mathcal{H}} \frac{|X_{p'} \cap H|}{|H|} \;\longrightarrow\; 1 - (1-p')^m \;\leq\; 1 - e^{-2Cs}.$$

So the deficiency cannot decay faster than exponentially in s, and $1 - 2^{-s}$ is sharp up to the constant C.

Looking ahead. Theorem 3.1 reduces the location of thresholds, up to a logarithm, to a purely combinatorial covering problem: decide whether $\mathcal{H}$ is p-small. To apply it one must show that *no* cover is cheap, which is a statement about all covers at once. The next chapter shows how linear programming duality turns it into the existence of a single object, a spread measure.

3.7 Notes and references

The conjecture was formulated by Kahn and Kalai in 2006 and published in [KK07], in the more general setting of product measures. Talagrand [Tal10] proposed the weaker fractional version discussed in Chapter 4, which was proved by Frankston, Kahn, Narayanan and Park [FKNP21], building on the breakthrough of Alweiss, Lovett, Wu and Zhang [ALWZ21] on the sunflower problem. The full conjecture was proved by Park and Pham [PP24a]; the proof presented here follows theirs, using binomial random sets $W_i \sim X_{cp}$, as in the lectures, instead of uniformly random sets of a fixed size. Corollary 3.7 is implicit in the same argument. Theorem 3.8 is [Pha25, Theorem 1.3]; it sharpens the theorem of Park and Pham [PP24b] on Talagrand's selector process conjecture, and its proof uses "towers" of minimum fragments.

4 Fractional Covers and Spread

To apply Theorem 3.1 one must show that $\mathcal{H}$ is *not* p-small, that is, that *every* cover of $\mathcal{H}$ has cost at least $\frac{1}{2}$. A universal statement about all covers is hard to verify directly. Linear programming duality converts it, after a relaxation, into an existential statement: the existence of a single well-spread probability measure on $\mathcal{H}$.

4.1 Fractional covers

Encode a cover $\mathcal{G}$ by its indicator function $g = \mathbf{1}_{\mathcal{G}} : 2^X \to \{0,1\}$. Then "$\mathcal{G}$ covers $\mathcal{H}$" reads $\sum_{G \subseteq H} g(G) \geq 1$ for all $H \in \mathcal{H}$, and $\mathrm{cost}_p(\mathcal{G}) = \sum_G g(G) p^{|G|}$. Hence

$$\begin{aligned} q(\mathcal{H}) &= \sup\Big\{p : \min_{g:2^X \to \{0,1\}} \Big\{\sum_G p^{|G|} g(G) : \sum_{G \subseteq H} g(G) \geq 1 \ \ \forall H \in \mathcal{H}\Big\} < \frac{1}{2}\Big\} \\ &\leq \sup\Big\{p : \min_{g:2^X \to [0,1]} \Big\{\sum_G p^{|G|} g(G) : \sum_{G \subseteq H} g(G) \geq 1 \ \ \forall H \in \mathcal{H}\Big\} < \frac{1}{2}\Big\} =: q_{\mathrm{f}}(\mathcal{H}). \end{aligned}$$

A function $g : 2^X \to [0,1]$ satisfying the constraints is a *fractional cover*; $\mathcal{H}$ is *weakly p-small* if it has a fractional cover of cost less than $\frac{1}{2}$; and $q_{\mathrm{f}}(\mathcal{H})$ is the *fractional expectation threshold* of Talagrand [Tal10].

A fractional cover still gives a first moment bound. If $X_p \in \langle \mathcal{H} \rangle$, say $X_p \supseteq H$, then $Z := \sum_{G \subseteq X_p} g(G) \geq \sum_{G \subseteq H} g(G) \geq 1$, so $\mathbb{P}(X_p \in \langle \mathcal{H} \rangle) \leq \mathbb{E} Z = \sum_G g(G) p^{|G|}$. Hence $q_{\mathrm{f}} \leq p_{\mathrm{c}}$, and with Theorem 3.1,

$$q(\mathcal{H}) \ \leq \ q_{\mathrm{f}}(\mathcal{H}) \ \leq \ p_{\mathrm{c}}(\mathcal{H}) \ \leq \ K\, q(\mathcal{H}) \log \ell(\mathcal{H}) \qquad (\ell(\mathcal{H}) \geq 2). \tag{4.1}$$

The weaker statement $p_{\mathrm{c}} \leq K q_{\mathrm{f}} \log \ell$ was conjectured by Talagrand and proved by Frankston, Kahn, Narayanan and Park [FKNP21].

4.2 LP duality and spread measures

For fixed p, the minimum over fractional covers is a linear program (the upper bound $g \leq 1$ can be dropped without changing the optimum):

$$\begin{aligned} &\text{(P)} \quad \text{minimise} \ \sum_{G \subseteq X} p^{|G|} g(G) \quad \text{subject to} \ \sum_{G \subseteq H} g(G) \geq 1 \ \ (H \in \mathcal{H}), \qquad g \geq 0; \\ &\text{(D)} \quad \text{maximise} \ \sum_{H \in \mathcal{H}} \lambda(H) \quad \text{subject to} \ \sum_{H \in \mathcal{H}:\ H \supseteq G} \lambda(H) \leq p^{|G|} \ \ (G \subseteq X), \quad \lambda \geq 0. \end{aligned}$$

By strong duality, (P) and (D) have the same optimal value. A feasible solution of (D) of value $\frac{1}{2}$ is, after multiplying by 2, a probability measure.

Proposition 4.1. *$\mathcal{H}$ is not weakly p-small if and only if there is a probability measure μ on $\mathcal{H}$ with*

$$\mu\big(\{H \in \mathcal{H} : \ H \supseteq G\}\big) \ \leq \ 2\, p^{|G|} \qquad \textit{for all } G \subseteq X.$$

Proof. If such μ exists, then $\lambda = \mu/2$ is feasible for (D) with value $\frac{1}{2}$, so the optimum of (P) is at least $\frac{1}{2}$. Conversely, if the optimum V of (P) is at least $\frac{1}{2}$, take an optimal λ^* for (D); then $\mu = \lambda^*/V$ is a probability measure with $\mu(\{H \supseteq G\}) \leq p^{|G|}/V \leq 2p^{|G|}$. □

This motivates the central definition. We use the normalisation customary in the literature, without the factor 2.

Definition 4.2 (Spread). A probability measure μ on $\mathcal{H} \subseteq 2^X$ is *q-spread* if

$$\mu\big(\{H \in \mathcal{H} : \ H \supseteq S\}\big) \ \le \ q^{|S|} \qquad \text{for every } S \subseteq X.$$

We say $\mathcal{H}$ is q-spread if it supports a q-spread measure. In words: no set S is contained in more than a $q^{|S|}$-fraction (in μ-measure) of the members of $\mathcal{H}$.

(In the lectures, "p-spread" was defined with the bound $2p^{|G|}$, which by Proposition 4.1 is exactly "not weakly p-small". The two conventions agree up to a factor 2 in q, since $2p^{|S|} \le (2p)^{|S|}$ for $|S| \ge 1$.)

Corollary 4.3. *If $\mathcal{H}$ supports a q-spread measure, then $\mathcal{H}$ is not (even weakly) q-small. Consequently, if all members of $\mathcal{H}$ have size at most ℓ,*

$$\mathbb{P}\big(X_{\min(1,\, 100\, q \log_2(2\ell))} \in \langle \mathcal{H} \rangle\big) \ \ge \ \frac{1}{2}.$$

Proof. There is a one-line direct argument. Let $\mathcal{G}$ be any cover of $\mathcal{H}$ and μ a q-spread measure. Then

$$\mathrm{cost}_q(\mathcal{G}) = \sum_{G \in \mathcal{G}} q^{|G|} \ \ge \ \sum_{G \in \mathcal{G}} \mu(\{H \supseteq G\}) \ = \ \mathbb{E}_{H \sim \mu}\big[\#\{G \in \mathcal{G} : G \subseteq H\}\big] \ \ge \ 1,$$

since every H contains at least one member of $\mathcal{G}$. The same computation applies to fractional covers. The second statement is Theorem 3.1. □

Spread measures as certificates. A spread measure is a *dual certificate*: it proves at once that no cover, however cleverly chosen, is cheap. Its existence is a local, checkable condition. In practice, *applying the Kahn–Kalai theorem means constructing a spread measure*; for symmetric families the uniform measure often works, and for structures inside a non-symmetric host (Chapter 7) constructing one is the main difficulty.

Example 4.4 *(Transversals).* In Example 2.12, let μ be uniform on transversals. If S is a partial transversal, then $\mu(\{H \supseteq S\}) = m^{-|S|}$; otherwise it is 0. So μ is $(1/m)$-spread and $\mathcal{H}$ is not $(1/m)$-small, as claimed.

Example 4.5 *(Hamilton paths).* Let μ be uniform on the $N!/2$ Hamilton paths of K_N and let S be a set of $s \ge 1$ edges. If S is not a linear forest, no Hamilton path contains it. Otherwise S consists of c vertex-disjoint paths covering v vertices, with $s = v - c$. Contracting each of these paths to a block, the Hamilton paths containing S correspond to the orderings of the $N - v + c = N - s$ blocks and remaining vertices, together with an orientation of each of the c blocks, up to reversal. Hence

$$\mu(\{H \supseteq S\}) = \frac{(N-s)!\, 2^c}{N!} \le 2^s \, \frac{(N-s)!}{N!} \le \Big(\frac{2e}{N}\Big)^s,$$

using that the product of the s largest factors of $N!$ is at least $(N!)^{s/N} \ge (N/e)^s$. Thus μ is $(2e/N)$-spread. So $\mathcal{H}$ is not $(2e/N)$-small (completing Example 2.13), and Corollary 4.3 gives $p_c(\text{Hamilton path}) = O(\log N/N)$. This is the correct order of magnitude, which historically required the work of Pósa, Komlós–Szemerédi and Bollobás [Pós76, KS83].

The same strategy gives, essentially for free, the threshold of a bounded-degree spanning tree T in $G(N,p)$ up to a constant, $p_c = \Theta(\log N/N)$ (the uniform measure on copies of T is $O(1/N)$-spread), recovering a theorem of Montgomery [Mon19]; and the thresholds for K_r-factors and for perfect matchings in random hypergraphs (Shamir's problem, Johansson–Kahn–Vu [JKV08]) up to a logarithmic factor.

4.3 Talagrand's conjecture: integral versus fractional

How much is lost in the relaxation $q(\mathcal{H}) \le q_{\mathrm{f}}(\mathcal{H})$?

Conjecture 4.6 (Talagrand [Tal10]). *There is an absolute constant K such that $q_{\mathrm{f}}(\mathcal{H}) \le K\, q(\mathcal{H})$ for every $\mathcal{H}$.*

If true, spread measures would characterise the expectation threshold up to a constant. The following was stated in the lectures.

Theorem 4.7. *There is an absolute constant C such that for every finite X and every $\mathcal{H} \subseteq 2^X$,*

$$q_{\mathrm{f}}(\mathcal{H}) \;\le\; C\, q(\mathcal{H}) \max\{1, \log\log|X|\}.$$

This follows by combining a rounding theorem of Pham [Pha25, Theorem 1.2] with a result of Fischer and Person [FP23, Theorem 10]. Pham's rounding theorem states that if $\mathcal{H}$ has a fractional cover of cost at most $\frac{1}{2}$ at density p supported on sets of size at most t, then $\mathcal{H}$ has an integral cover of cost at most $\frac{1}{2}$ at density $cp/\log t$. Very recently Park [Par26] replaced $\log\log|X|$ by $\log\log(1/q(\mathcal{H}))$, a bound independent of the size of the ground set.

Looking ahead. From now on, applying Theorem 3.1 means exhibiting a spread measure. The next three chapters do this in increasingly demanding settings: for sunflowers, where the uniform measure on a well-chosen subfamily works (Chapter 5); for powers of Hamilton cycles, where a smoother form of spread removes the logarithm (Chapter 6); and inside a dense host graph, where the measure has to be built by hand (Chapter 7).

4.4 Notes and references

The fractional expectation threshold and the conjecture $q_{\mathrm{f}}(\mathcal{H}) \le Kq(\mathcal{H})$ are from Talagrand's paper [Tal10], which also contains the conjecture $p_{\mathrm{c}} \le Kq_{\mathrm{f}} \log \ell$ proved in [FKNP21]. The duality between fractional covers and spread measures is standard linear programming duality (Appendix A.3); the normalisation of Definition 4.2 is that of [FKNP21, PSSS22, Spi23]. For hypergraphs, [FKNP21] and the sunflower papers [ALWZ21, Rao20] write instead that $\mathcal{H}$ is *κ-spread* (or R-spread) if $|\{H \in \mathcal{H} : H \supseteq S\}| \le \kappa^{-|S|}|\mathcal{H}|$; this is the uniform measure being κ^{-1}-spread in the sense of Definition 4.2. Theorem 4.7 combines the rounding theorem of Pham [Pha25, Theorem 1.2] with [FP23, Theorem 10]; see the discussion in Park's paper [Par26], which improves the factor to $\log\log(1/q(\mathcal{H}))$. Further special cases of Talagrand's conjecture are treated by Fischer and Person [FP23]. Spanning trees in $G(N,p)$ are due to Montgomery [Mon19], and K_r-factors and hypergraph perfect matchings to Johansson, Kahn and Vu [JKV08]; deriving such results, up to the logarithm, from spread measures was one of the motivations of [KK07, FKNP21].

5 The Sunflower Problem

A family of sets is a sunflower if all pairwise intersections coincide. How large can a family of k-sets be without containing a sunflower with r petals? This classical question of Erdős and Rado saw dramatic progress in 2019, and its solution is closely intertwined with the Kahn–Kalai story: the ideas behind the recent bounds led to the proof of Talagrand's fractional conjecture, and conversely the best bound is now a short consequence of the Park–Pham theorem and spread measures.

5.1 Sunflowers

Definition 5.1 (Sunflower). Sets $S_1, \dots, S_r$ form an *r-sunflower* if they are distinct and have a common pairwise intersection: $S_i \cap S_j = Y$ for all $i \neq j$. The set Y is the *kernel* and the sets $S_i \setminus Y$, which are pairwise disjoint, are the *petals*.

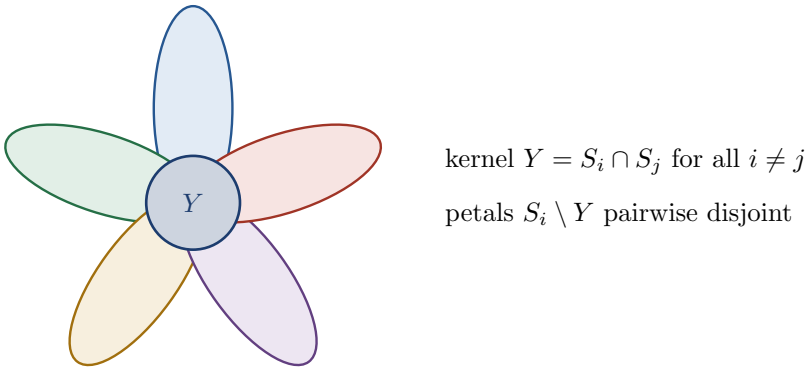


Figure 5.1: A 5-sunflower.

The Erdős–Rado sunflower lemma [ER60] states that any family of more than $k!\,(r-1)^k$ sets of size k contains an r-sunflower, and Erdős and Rado conjectured that $(C_r)^k$ sets suffice. After decades of little progress, Alweiss, Lovett, Wu and Zhang [ALWZ21] improved the bound to roughly $(\log k)^k$ for fixed r. Their argument was simplified and sharpened by Rao [Rao20], who obtained $(Cr\log(rk))^k$ and introduced the language of spread families, and then by Bell, Chueluecha and Warnke [BCW21]. The current best bound has the following short proof via spread measures and Theorem 3.1.

5.2 A proof via spread measures

Theorem 5.2 ([BCW21]). *There is an absolute constant C such that for all integers $k, r \geq 2$, every family of more than $(Cr\log k)^k$ distinct sets of size k contains an r-sunflower.*

Proof. Let $L := 400\,r\log_2 k$ and $p := 1/L$, and let $\mathcal{H}$ be a family of more than L^k distinct k-sets. For $G \subseteq X$ let $\mathcal{H}_G := \{H \setminus G : \ H \in \mathcal{H},\ H \supseteq G\}$ be the *link* of G.

Step 1: reduction to the spread case. Choose $G \subseteq X$ maximising $|\mathcal{H}_G|\,L^{|G|}$ (the empty set is allowed). Then $|\mathcal{H}_G|\,L^{|G|} \geq |\mathcal{H}| > L^k$. Since $|\mathcal{H}_G| \leq 1$ when $|G| = k$, this forces $|G| < k$, so $\mathcal{H}_G$ is a nonempty family of nonempty $(k-|G|)$-sets, all disjoint from G. By the maximality of G, for every nonempty S disjoint from G,

$$\#\{H' \in \mathcal{H}_G : \ H' \supseteq S\} = |\mathcal{H}_{G\cup S}| \ \leq \ L^{-|S|}\,|\mathcal{H}_G| = p^{|S|}\,|\mathcal{H}_G|.$$

Hence the uniform measure on $\mathcal{H}_G$ is p-spread. In words, if the family were too concentrated on some set S, we would move S into the potential kernel G; this is the step described in the lectures as "put G in the potential core, and argue in $H \setminus G$".

Step 2: a random colouring. Since the members of $\mathcal{H}_G$ have size at most k and $\log_2(2k) \le 2\log_2 k$, Corollary 4.3 gives $\mathbb{P}(X_{p'} \in \langle \mathcal{H}_G \rangle) \ge \frac{1}{2}$ for $p' = 200p\log_2 k = \frac{1}{2r}$. Colour every element of X independently and uniformly with one of $2r$ colours. Each colour class $X^{(i)}$ is distributed as $X_{1/(2r)} = X_{p'}$, so

$$\mathbb{E}\big[\#\{i \le 2r : \ X^{(i)} \in \langle \mathcal{H}_G \rangle\}\big] \ \ge \ 2r \cdot \tfrac{1}{2} = r.$$

Fix a colouring in which at least r colour classes contain a member of $\mathcal{H}_G$, and pick $H'_1, \dots, H'_r \in \mathcal{H}_G$ in distinct colour classes. They are pairwise disjoint and nonempty, hence distinct, so $H'_1 \cup G, \dots, H'_r \cup G \in \mathcal{H}$ form an r-sunflower with kernel G. This proves the theorem with $C = 400/\log 2$. □

Remark 5.3. The proof uses Theorem 3.1 only through Corollary 4.3 for *uniform* measures on *uniform* families. In this special case the statement "a q-spread k-uniform family is hit by $X_{Cq\log k}$ with probability $\ge \frac{1}{2}$" is the *spread lemma* of [ALWZ21, Rao20, BCW21], whose proof already contains the germ of the encoding argument of Lemma 3.4. The logical order in the lectures, Kahn–Kalai first and sunflowers second, is the reverse of the historical one.

Looking ahead. In the sunflower problem the factor $\log k$ coming from Theorem 3.1 is affordable, and in general it cannot be avoided (Example 2.12). For many spanning structures, however, the true threshold carries no logarithm at all. The next chapter shows how a finer spread condition recovers such sharp results.

5.3 Notes and references

The sunflower lemma and the sunflower conjecture are due to Erdős and Rado [ER60]. The bound of roughly $(\log k)^k$ for fixed r is the breakthrough of Alweiss, Lovett, Wu and Zhang [ALWZ21]. Rao [Rao20] gave a short proof via a coding argument, obtaining $(Cr\log(rk))^k$ and introducing the terminology of spread families, and Bell, Chueluecha and Warnke [BCW21] removed r from the logarithm, proving Theorem 5.2. The ideas of [ALWZ21] were the starting point of the proof of Talagrand's fractional conjecture [FKNP21], and through it of [PP24a].

6 Smoother Spread: Removing the Logarithm

The logarithmic factor in the Park–Pham theorem is the price of halving the minimum fragments only once per round. In this chapter we see where exactly it comes from in the proof, and how a finer spread condition, due to Spiro, removes it.

6.1 Where the logarithm comes from

The factor $\log \ell$ in Theorem 3.1 is sometimes necessary (Examples 2.12 and 2.13), but often it is not. Where does it come from in the proof? The minimum fragments only *halve* in size from one round to the next, so $\log_2 \ell$ rounds are needed. If one could guarantee that in each round the fragments shrink by a *polynomial* factor, say from r_i to $r_{i+1} \le r_i N^{-\alpha}$, then $O(1/\alpha)$ rounds would suffice and the logarithm would disappear.

Shrinking is governed by how a random member $H \sim \mu$ intersects a fixed set A (think of A as the current fragment). A q-spread measure controls such intersections only through a union bound,

$$\mu\big(|H \cap A| \ge t\big) \;\le\; \sum_{S \in \binom{A}{t}} \mu(H \supseteq S) \;\le\; \binom{|A|}{t} q^t \;\le\; \Big(\frac{e|A|q}{t}\Big)^t,$$

which is useless unless $t \gtrsim |A|q$. The binomial factor is the price of ignoring that for most t-subsets $S \subseteq A$ the probability $\mu(H \supseteq S)$ is much smaller than q^t. Spiro's notion of spread asks for the intersection bound directly.

6.2 Smoother spread

Definition 6.1 (Smoother spread [Spi23]). Let $r_1 > r_2 > \cdots > r_m = 1$ be integers. A probability measure μ on $\mathcal{H}$ is $(q; r_1, \dots, r_m)$-*spread* if every $H \in \operatorname{supp} \mu$ has $|H| \le r_1$, and for every $1 \le i < m$, every $A \subseteq X$ with $|A| \le r_i$ and every integer $t \ge r_{i+1}$,

$$\mu\big(\{H \in \mathcal{H} :\; |H \cap A| \ge t\}\big) \;\le\; q^t.$$

Thus sets of size at most r_i rarely meet a random member of $\mathcal{H}$ in r_{i+1} or more elements, with no binomial loss.

Theorem 6.2 (Spiro [Spi23]). *There is an absolute constant K such that the following holds. If $\mathcal{H}$ is r_1-uniform and the uniform measure on $\mathcal{H}$ is $(q; r_1, \dots, r_m)$-spread, then*

$$\mathbb{P}\big(X_{\min(1, Kmq)} \in \langle \mathcal{H} \rangle\big) \;\ge\; \frac{1}{2}.$$

(Spiro states the result for a uniformly random subset of X of size $Kmq|X|$, with a probability tending to 1 as K grows. The binomial version follows by a standard coupling, and the lectures stated it for general measures μ.)

Proof sketch. Take $W_1, \dots, W_m$ independent copies of X_{cq} and run the minimum fragment iteration of Chapter 3, but with thresholds $r_2, r_3, \dots$ instead of $\ell/2, \ell/4, \dots$. After $W_1, \dots, W_i$ have been exposed, almost all $H \in \mathcal{H}$ (in μ-measure) give rise to a fragment of size at most r_{i+1}. The members whose fragment is larger are handled by an encoding argument as in Lemma 3.4, in which the binomial factor $\binom{\ell}{t}$ is replaced by the intersection bound of Definition 6.1 applied with A the fragment from the previous round. Since $r_m = 1$, after m rounds the fragments are empty, and the total density used is mcq. □

6.3 Powers of Hamilton cycles

Example 6.3 *(Powers of Hamilton cycles).* Let C_N^k be the k-th power of the N-cycle: vertices $i, j \in \mathbb{Z}_N$ are adjacent when their cyclic distance is at most k. Let $\mathcal{H}$ be the family of copies of C_N^k in K_N, a kN-uniform family. There are $N!/(2N)$ copies, so the naive bound $\mathrm{cost}_p(\mathcal{H}) \le N!\, p^{kN} \le (Np^k)^N$ shows $q(\mathcal{H}) \ge cN^{-1/k}$, and the uniform measure is $O(N^{-1/k})$-spread, so $q(\mathcal{H}) = \Theta(N^{-1/k})$. Theorem 3.1 therefore gives $p_c(\mathcal{H}) = O(N^{-1/k} \log N)$.

It was stated in the lectures that for $k \ge 2$ the uniform measure is in fact $(q; r_1, \dots, r_m)$-spread with $q = \Theta_k(N^{-1/k})$, $r_1 = kN$ and $r_{i+1} \le r_i N^{-\alpha_k}$ for some $\alpha_k > 0$, so that $m = O_k(1)$. Theorem 6.2 then gives the optimal bound

$$p_c(\mathcal{H}) = \Theta\big(N^{-1/k}\big) \qquad (k \ge 2).$$

For $k \ge 3$ this is due to Riordan [Rio00] via the second moment method. For $k = 2$, where the second moment method fails, it is a theorem of Kahn, Narayanan and Park [KNP21]. Spiro observed that the square of a Hamilton cycle is $(CN^{-1/2}; 2N, C_0 N^{1/2}, 1)$-spread, so the case $m = 3$ of Theorem 6.2 recovers [KNP21].

Remark 6.4 *(Why $k \ge 2$? A heuristic).* For $k = 1$ (Hamilton cycles) the threshold is $\sim \log N/N$, so no such smooth spread with $m = O(1)$ can exist. Here is one way to see the difference. Let A be a fixed copy of C_N^k and H a uniformly random copy. Each edge of A lies in H with probability about $2kN/N^2 = 2k/N$, so $|H \cap A|$ is roughly Poisson with constant mean $\approx 2k^2$, and $\mu(|H \cap A| \ge t) \approx (O_k(1)/t)^t$. For this to be at most q^t we need $t \gtrsim 1/q$.

- For $k = 1$, $q = \Theta(1/N)$ forces $t \gtrsim N$: there is no room to shrink $r_1 = N$.
- For $k \ge 2$, $q = \Theta(N^{-1/k})$ is much larger than the typical per-edge probability $\Theta(1/N)$, and $t \gtrsim N^{1/k}$ suffices: fragments can drop from size kN to size about $N^{1/k}$ in one round.

The slack between q and the typical behaviour is what the smooth spread condition exploits.

Looking ahead. So far the host structure has been the complete graph K_N, and its symmetry made the uniform measure on copies a natural candidate. In the next chapter K_N is replaced by an arbitrary dense graph, and the spread measure has to be constructed.

6.4 Notes and references

The notion of $(q; r_1, \dots, r_m)$-spread and Theorem 6.2 are from Spiro [Spi23], who formulates the condition for sets A with $r_{i+1} \le |A| \le r_i$; this is equivalent to Definition 6.1, since the condition is vacuous for smaller A. A special case was used implicitly by Kahn, Narayanan and Park [KNP21] to determine the threshold for the square of a Hamilton cycle. For $k \ge 3$ the threshold for the k-th power of a Hamilton cycle was determined by Riordan [Rio00] with the second moment method.

Part III

Beyond the Complete Graph

7 Robust Thresholds

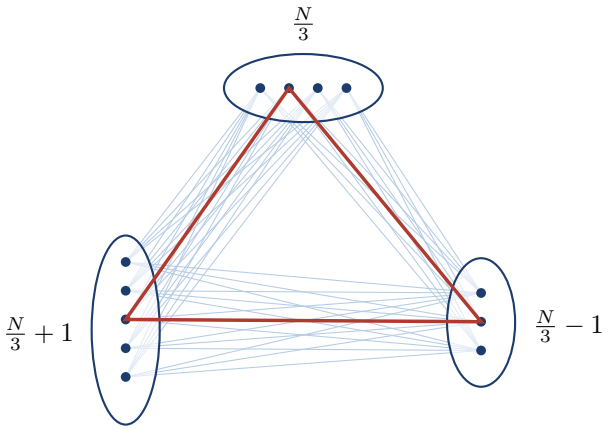
So far the host has always been the complete graph K_N, and the symmetry of K_N made the uniform measure on copies of H a natural spread measure. We now replace K_N by an arbitrary dense graph G. The resulting "robust" theorems interpolate between two classical worlds, extremal graph theory (the case $p = 1$) and random graphs (the case $G = K_N$), and the spread method of Chapter 4 is what makes them accessible.

7.1 Two classical worlds

Dirac-type theorems. Dirac's theorem says that a graph on $N \geq 3$ vertices with minimum degree $\delta(G) \geq N/2$ is Hamiltonian [Dir52]. Among its many generalisations:

Theorem 7.1 (Hajnal–Szemerédi [HS70]). *If $(k+1) \mid N$ and $\delta(G) \geq \left(1 - \frac{1}{k+1}\right)N$, then G contains a K_{k+1}-factor.*

The degree condition is tight (Figure 7.1). Take the complete $(k+1)$-partite graph with one part of size $\frac{N}{k+1} + 1$, one of size $\frac{N}{k+1} - 1$, and the others of size $\frac{N}{k+1}$. Every copy of K_{k+1} uses exactly one vertex from each part, so a K_{k+1}-factor would need all parts to have the same size; yet the minimum degree is $\left(1 - \frac{1}{k+1}\right)N - 1$. (The case $k = 2$ is the earlier Corrádi–Hajnal theorem [CH63].)



Complete tripartite graph with unbalanced parts: $\delta(G) = \frac{2}{3}N - 1$. Every triangle (red) takes one vertex from each part, so there is no triangle factor.

Figure 7.1: Tightness of the Hajnal–Szemerédi theorem for $k = 2$.

Theorem 7.2 (Komlós–Sárközy–Szemerédi [KSS98b]). *For N sufficiently large, if $\delta(G) \geq \left(1 - \frac{1}{k+1}\right)N$ then G contains the k-th power of a Hamilton cycle. (This is the Pósa–Seymour conjecture for large N.)*

Theorem 7.3 (Komlós–Sárközy–Szemerédi [KSS95]). *For every $\gamma > 0$ and Δ there is N_0 such that if $N \geq N_0$ and $\delta(G) \geq (\frac{1}{2} + \gamma)N$, then G contains every N-vertex tree T with $\Delta(T) \leq \Delta$.*

Random analogues. The corresponding questions in $G(N,p)$ ask for thresholds.

Theorem 7.4 (Johansson–Kahn–Vu [JKV08]). $\mathbb{P}\big(G(N,p)$ *contains a K_{k+1}-factor*$\big) > \frac{1}{2}$ *for $p \geq C\,(\log N)^{1/\binom{k+1}{2}} N^{-2/(k+1)}$, and this is optimal up to the value of C.*

Theorem 7.5 (Montgomery [Mon19]). *Let T be an N-vertex tree with $\Delta(T) \leq \Delta$. Then $G(N,p)$ contains a copy of T with probability greater than $\frac{1}{2}$ for $p \geq C \log N/N$, where $C = C(\Delta)$.*

7.2 Robust versions

Definition 7.6. For a graph G, let G_p be the random subgraph of G in which each edge is kept independently with probability p. (In the language of Chapter 2, $G_p = X_p$ with $X = E(G)$.)

A *robust* version of a Dirac-type theorem asks: if $\delta(G)$ satisfies the extremal condition, how small can p be so that G_p still contains the spanning structure w.h.p.? The answer cannot be better than for $G = K_N$, and the remarkable fact is that it is usually the same. The first result of this kind is due to Krivelevich, Lee and Sudakov [KLS14]: if $\delta(G) \geq N/2$ then G_p is Hamiltonian w.h.p. for $p \geq C \log N/N$.

Theorem 7.7 (Allen et al. [ABC+24] for $k = 2$; Pham–Sah–Sawhney–Simkin [PSSS22] for all k). *Let $\delta(G) \geq \left(1 - \frac{1}{k+1}\right)N$, with $(k+1) \mid N$.*

(a) G_p contains a K_{k+1}-factor w.h.p. (in particular with probability at least $\frac{1}{2}$) whenever

$$p \;\geq\; C\,(\log N)^{1/\binom{k+1}{2}}\, N^{-2/(k+1)}.$$

(b) The number of K_{k+1}-factors in G is at least $(cN)^{(1-\frac{1}{k+1})N}$.

Theorem 7.8 (Pham–Sah–Sawhney–Simkin [PSSS22]). *For every $\gamma > 0$ and Δ there is C such that if $\delta(G) \geq (\frac{1}{2} + \gamma)N$ and T is an N-vertex tree with $\Delta(T) \leq \Delta$, then G_p contains a copy of T w.h.p. for $p \geq C \log N/N$.*

Part (b) of Theorem 7.7 is the counting statement that appears next to the robust threshold in the lectures. It says that G has, up to the value of c, as many K_{k+1}-factors as K_N itself, since K_N has $N!/\big(((k+1)!)^{N/(k+1)}(\frac{N}{k+1})!\big) = (\Theta(N))^{(1-\frac{1}{k+1})N}$ of them. We will see in Remark 7.17 that it is a free by-product of the spread approach.

7.3 The method: regularity and a spread blow-up lemma

By Corollary 4.3, to prove a robust threshold it suffices to construct a spread measure on copies of H inside G. Without the symmetry of K_N one has to build it by hand, and the tool is Szemerédi's regularity method.

For disjoint $A, B \subseteq V(G)$ write $d(A,B) = e_G(A,B)/(|A||B|)$ for the edge density.

Definition 7.9 (Regular and super-regular pairs). The pair (A,B) is *ε-regular* if $|d(X,Y) - d(A,B)| \leq \varepsilon$ for all $X \subseteq A$, $Y \subseteq B$ with $|X| \geq \varepsilon|A|$ and $|Y| \geq \varepsilon|B|$. It is *(ε,δ)-super-regular* if moreover every $a \in A$ has at least $\delta|B|$ neighbours in B and every $b \in B$ has at least $\delta|A|$ neighbours in A.

Theorem 7.10 (Szemerédi's regularity lemma [Sze78]). *For every $\varepsilon > 0$ there is $M = M(\varepsilon)$ such that every graph G admits a partition $V(G) = V_1 \cup \cdots \cup V_m$ with $m \leq M$ and $\big||V_i| - |V_j|\big| \leq 1$, such that all but at most εm^2 of the pairs (V_i, V_j) are ε-regular.*

A regular pair can always be cleaned up into a super-regular one.

Lemma 7.11. *Let $0 < \varepsilon \leq \frac{1}{2}$. If (A,B) is ε-regular with density d and $|A| = |B| = n$, then there are $A' \subseteq A$, $B' \subseteq B$ with $|A'|, |B'| \geq (1-\varepsilon)n$ such that (A', B') is $(2\varepsilon, d - 2\varepsilon)$-super-regular.*

Proof. Let $A_0 = \{a \in A : \deg(a,B) < (d-\varepsilon)n\}$. If $|A_0| \geq \varepsilon n$ then $d(A_0, B) < d - \varepsilon$, contradicting regularity; so $|A_0| < \varepsilon n$, and similarly for $B_0 = \{b \in B : \deg(b,A) < (d-\varepsilon)n\}$. Let $A' = A \setminus A_0$ and $B' = B \setminus B_0$. Every $a \in A'$ has at least $(d-\varepsilon)n - |B_0| \geq (d-2\varepsilon)n \geq (d-2\varepsilon)|B'|$ neighbours in B', and symmetrically. If $X \subseteq A'$, $Y \subseteq B'$ with $|X| \geq 2\varepsilon|A'| \geq \varepsilon n$ and $|Y| \geq 2\varepsilon|B'| \geq \varepsilon n$, then $|d(X,Y) - d| \leq \varepsilon$ and $|d(A',B') - d| \leq \varepsilon$, so $|d(X,Y) - d(A',B')| \leq 2\varepsilon$. □

The blow-up lemma says that, for embedding bounded-degree graphs, super-regular pairs behave like complete bipartite graphs.

Theorem 7.12 (Blow-up lemma, Komlós–Sárközy–Szemerédi [KSS97]). *For all $r, \Delta \in \mathbb{N}$ and $\delta > 0$ there is $\varepsilon > 0$ such that the following holds. Let R be a graph on $[r]$, and let G be a graph on vertex sets $V_1, \dots, V_r$ with $|V_i| = n$, such that $G[V_i, V_j]$ is (ε, δ)-super-regular for every edge $ij \in R$. Let $\overline{G}$ be obtained by putting a complete bipartite graph between V_i and V_j for every $ij \in R$. If H has $\Delta(H) \le \Delta$ and $H \subseteq \overline{G}$, then $H \subseteq G$.*

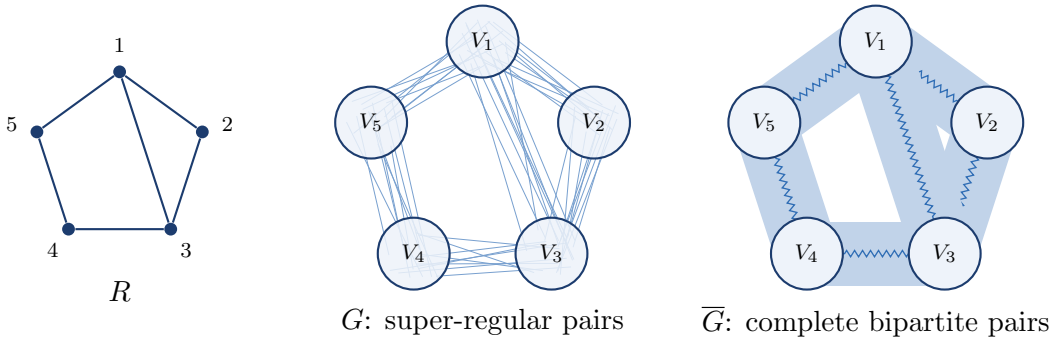


Figure 7.2: The blow-up lemma. Each vertex of the reduced graph R is blown up to a cluster of size n; edges of R become super-regular pairs in G (middle) or complete bipartite graphs in $\overline{G}$ (right). Bounded-degree spanning graphs that embed in $\overline{G}$ also embed in G.

For thresholds we need not just one embedding, but a *well-distributed random* embedding.

Definition 7.13 (Vertex-spread). A probability measure on injections $\phi : V(H) \to V(G)$ is *q-vertex-spread* if for all k, all distinct $h_1, \dots, h_k \in V(H)$ and all $g_1, \dots, g_k \in V(G)$,

$$\mathbb{P}\big(\phi(h_i) = g_i \text{ for } i = 1, \dots, k\big) \;\le\; q^k.$$

A uniformly random bijection $V(H) \to [N]$ is (e/N)-vertex-spread, which is the property of K_N we used implicitly in Chapter 4.

Theorem 7.14 (Spread blow-up lemma; Nenadov–Pham [NP24]). *In the setting of Theorem 7.12, there is a probability measure on embeddings $\phi : V(H) \hookrightarrow V(G)$ of H into G which is (C/n)-vertex-spread, where $C = C(r, \Delta, \delta)$.*

This combines the spread perfect matchings in super-regular pairs constructed in [PSSS22] with the randomised, algorithmic version of the blow-up lemma [KSS98a]. See [NP24, Lemma 2] for the precise formulation, which also allows the images of a few vertices to be restricted.

7.4 From vertex-spread to edge-spread

Lemma 7.15. *Let G be a graph on $N \le Ln$ vertices and H a graph on $N_H \le N$ vertices with $\Delta(H) \le \Delta$, and let ϕ be drawn from a (C/n)-vertex-spread measure on embeddings of H into G. Let μ be the law of the edge set $\phi(H) \subseteq E(G)$. If $S \subseteq E(G)$ spans $v(S)$ vertices and has $c(S)$ components, then*

$$\mu\big(\phi(H) \supseteq S\big) \;\le\; (CL\Delta)^{v(S)}\, n^{-(v(S)-c(S))}.$$

Proof. If $\phi(H) \supseteq S$, the vertices of S have preimages in H, and each component of S is the image of a connected subgraph of H. Fix a spanning tree of each component of S and a root. The preimage of a root can be chosen in at most $N_H \le Ln$ ways, and then the preimage of every further vertex, which is adjacent along the tree to an already placed vertex, in at most Δ ways. So there are at most $(Ln)^{c(S)}\Delta^{v(S)}$ candidate preimage assignments, each realised with probability at most $(C/n)^{v(S)}$. □

Each component of S is (the image of) a subgraph $F \subseteq H$, and $v(F) - 1 \geq e(F)/m_1(H)$, where $m_1(H) := \max\{e(F)/(v(F)-1) : F \subseteq H,\ v(F) \geq 2\}$. Summing over components gives $v(S) - c(S) \geq |S|/m_1(H)$. Together with $v(S) \leq 2|S|$, this shows that μ is $O(n^{-1/m_1(H)})$-spread, and Corollary 4.3 yields:

Corollary 7.16. *In the setting of Lemma 7.15, G_p contains a copy of H with probability at least $\frac{1}{2}$ for $p \geq K\, n^{-1/m_1(H)} \log N$, with K depending on C, L, Δ.*

- For trees $m_1 = 1$, giving $p = O(\log N/N)$, the bound of Theorem 7.8. The hard part is to construct the vertex-spread measure from the minimum degree condition, using regularity, the spread blow-up lemma, and a careful treatment of exceptional vertices.
- For K_{k+1}-factors $m_1 = \frac{k+1}{2}$, giving $p = O(N^{-2/(k+1)} \log N)$. This is off from Theorem 7.7(a) by a factor $(\log N)^{1-1/\binom{k+1}{2}}$; removing it requires additional ideas.
- For powers of Hamilton cycles one combines Theorem 7.14 with Spiro's smooth spread theorem (Theorem 6.2). This is how Nenadov and Pham [NP24] prove an approximate version of a conjecture of Böttcher, Parczyk, Sgueglia and Skokan on powers of Hamilton cycles in randomly perturbed graphs.

Remark 7.17 *(Spread implies counting).* Let H be a K_{k+1}-factor on N vertices and suppose G supports a (C/N)-vertex-spread measure on embeddings of H. A fixed K_{k+1}-factor F of G is the image of exactly $|\operatorname{Aut}(H)|$ embeddings, each of probability at most $(C/N)^N$. Hence $\mu(\{F\}) \leq |\operatorname{Aut}(H)|(C/N)^N$, and the number of K_{k+1}-factors of G is at least

$$\frac{1}{\max_F \mu(\{F\})} \;\geq\; \frac{(N/C)^N}{((k+1)!)^{N/(k+1)}\,(\frac{N}{k+1})!} \;\geq\; (c\,N)^{(1-\frac{1}{k+1})N},$$

which is Theorem 7.7(b). Spread measures are "entropy certificates": a measure that is not concentrated anywhere must have a large support.

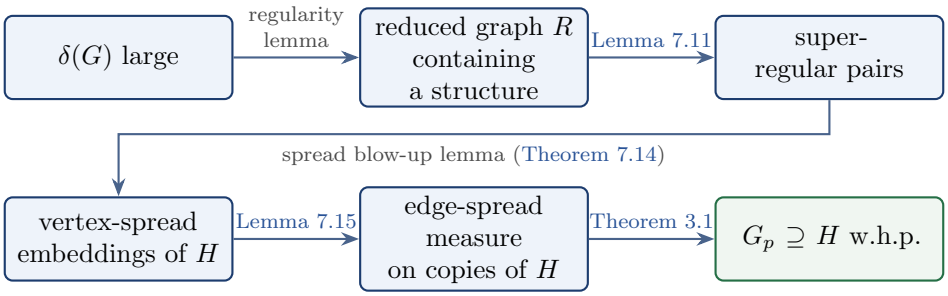


Figure 7.3: The pipeline for robust thresholds.

Looking ahead. In every application so far the work went into showing that *no* first moment obstruction exists, by exhibiting a spread measure. The last chapter goes in the opposite direction: there the naive first moment fails badly, and the task is to *construct* a good first moment obstruction.

7.5 Notes and references

Dirac's theorem is from [Dir52], the Corrádi–Hajnal and Hajnal–Szemerédi theorems from [CH63, HS70], and the Komlós–Sárközy–Szemerédi theorems from [KSS95, KSS98b]. The random analogues are due to Johansson, Kahn and Vu [JKV08] and Montgomery [Mon19]. Robust versions of Dirac-type theorems were initiated by Krivelevich, Lee and Sudakov [KLS14];

the robust Corrádi–Hajnal theorem is due to Allen, Böttcher, Corsten, Davies, Jenssen, Morris, Roberts and Skokan [ABC+24], and the general toolkit, including Theorems 7.7 and 7.8 and spread perfect matchings in super-regular pairs, to Pham, Sah, Sawhney and Simkin [PSSS22]. The regularity lemma is from [Sze78], the blow-up lemma from [KSS97] and its algorithmic version from [KSS98a]. The spread blow-up lemma (Theorem 7.14) and its application to powers of Hamilton cycles in randomly perturbed graphs are due to Nenadov and Pham [NP24]. Lemma 7.15 and Remark 7.17 are written out here in the generality needed for these notes.

8 Random Cayley Graphs and First Moment Obstructions

The Park–Pham theorem says that thresholds are governed by first moment obstructions. Applying it in the upper-bound direction requires showing that *no* cheap cover exists (Chapters 4 to 7). In this last chapter we look in the opposite direction: to show that a random structure does *not* contain something, we must *find* a cheap cover. In random Cayley graphs the naive first moment fails spectacularly, and finding good first moment obstructions becomes a problem in additive combinatorics.

8.1 Setup

Let G be a finite abelian group of order N, written additively (for example $G = \mathbb{Z}_N$ or $G = \mathbb{F}_q^d$). For a symmetric set $S = -S \subseteq G \setminus \{0\}$, the *Cayley graph* $\mathrm{Cay}(G, S)$ has vertex set G, with $x \sim y$ if and only if $x - y \in S$; it is $|S|$-regular.

Definition 8.1 (Random Cayley graph). The *random Cayley graph* $G(p)$ is $\mathrm{Cay}(G, S)$, where for each pair $\{g, -g\}$ with $g \neq 0$, independently, both g and $-g$ are put into S with probability p.

So $G(p)$ is a random graph of degree about pN, like $G(N, p)$, but it is generated by only about $N/2$ independent coins instead of $\binom{N}{2}$. Its edges are therefore highly correlated. A basic question is the size of its largest independent set. Since any graph of maximum degree d has an independent set of size at least $N/(d+1)$, we always have $\alpha(G(p)) \geq (1 - o(1))/p$ when $pN \to \infty$; for comparison, $\alpha(G(N, p)) \approx 2\log(Np)/p$.

8.2 Difference sets and the first moment

For $A \subseteq G$ let $A - A := \{a_2 - a_1 : a_1, a_2 \in A\}$. A set A is independent in $\mathrm{Cay}(G, S)$ if and only if no difference of two elements of A lies in S:

$$A \text{ independent} \iff (A - A) \setminus \{0\} \subseteq G \setminus S.$$

Hence, writing $\pi(D)$ for the number of pairs $\{g, -g\}$ meeting a symmetric set D, so that $|D|/2 \leq \pi(D) \leq |D|$,

$$\mathbb{E}\big[N_{n\text{-indep}}(G(p))\big] = \sum_{|A|=n} (1-p)^{\pi((A-A)\setminus\{0\})},$$

which is $\sum_{|A|=n}(1-p)^{|A-A|}$ up to a factor of 2 in the exponent. So *controlling the first moment amounts to enumerating sets A with small difference set $|A - A|$*, a classical theme in additive combinatorics [Gre05].

8.3 The naive first moment fails

Take $G = \mathbb{Z}_N$ and let A be any n-subset of the interval $\{0, 1, \ldots, 6n-1\}$ (Figure 8.1). Then $A - A \subseteq \{-(6n-1), \ldots, 6n-1\}$, so $(1-p)^{\pi((A-A)\setminus\{0\})} \geq (1-p)^{12n}$, and there are $\binom{6n}{n} \geq 2^n$ such sets. Therefore, for $p \leq \frac{1}{100}$,

$$\mathbb{E}\big[N_{n\text{-indep}}(G(p))\big] \;\geq\; 2^n(1-p)^{12n} \;\geq\; \big(2e^{-24p}\big)^n \;\geq\; 1.5^n,$$

for every $n \le N/6$. Yet, as we will see, $\alpha(G(p))$ is much smaller than N whenever $p \gg N^{-1/2}\log N$. The expectation is useless.

This is exactly the phenomenon of Example 2.6, in a far more extreme form. The event that S avoids the short interval $\{\pm 1, \dots, \pm(6n-1)\}$ has probability about $(1-p)^{6n}$, which is tiny when $pn \to \infty$; but when it happens, all $\binom{6n}{n}$ subsets of the interval become independent simultaneously. To fix this we need covers: small, cleverly chosen sets $F \subseteq A - A$, with few possible F's, each of which is unlikely to avoid S.

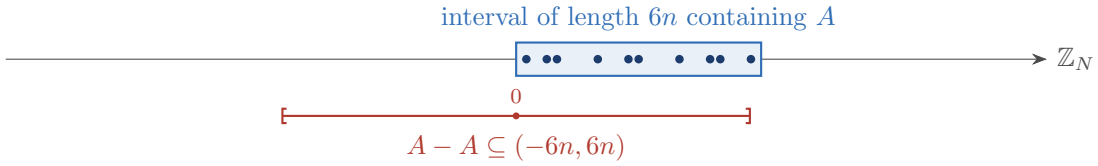


Figure 8.1: Sets inside a short interval have small difference sets. There are many of them, and they are all independent as soon as S avoids the short interval of differences.

8.4 What is known

The interesting regime is $p = o(1)$, in particular $p = N^{-c}$ with $c \in (0, \frac{1}{2})$, where the dependence on p matters.

Theorem 8.2 (Alon [Alo07, Alo13]). *With high probability,* $\alpha(G(p)) = O\big(p^{-2}\log^2 N\big)$.

For constant p this gives $O(\log^2 N)$. Conlon, Fox, Pham and Yepremyan [CFPY24] improved it to $O(\log N \log\log N)$ for every abelian group, which is best possible for some groups, including $\mathbb{F}_2^d$ [Gre05]. In the sparse regime the exponent of p is the main question.

Conjecture 8.3 (Alon [Alo13]). *With high probability,* $\alpha(G(p)) = O\big(p^{-1}(\log N)^{O(1)}\big)$.

This would match the lower bound $\Omega(1/p)$ up to logarithmic factors, so the random Cayley graph would have essentially the same independence number as a random graph of the same density.

Theorem 8.4 (Alon–Pham [AP25]). *Let* $p \le \frac{1}{2}$. *With high probability,*

$$\alpha(G(p)) = O\big(p^{-3/2}(\log N)^{O(1)}\big).$$

This was the first improvement of the exponent of p in Theorem 8.2. We prove Theorem 8.2 and then Theorem 8.4; both proofs construct first moment obstructions.

8.5 Alon's bound via a first moment obstruction

Let $\mathcal{H}_n := \{(A-A)\setminus\{0\} : A \subseteq G,\ |A| = n\}$. The event $\alpha(G(p)) \ge n$ is the event that $G \setminus S$ contains a member of $\mathcal{H}_n$. For a family $\mathcal{G}$ of symmetric subsets of $G\setminus\{0\}$ covering $\mathcal{H}_n$, the union bound, exactly as in (2.2), gives

$$\mathbb{P}\big(\alpha(G(p)) \ge n\big) \ \le\ \sum_{F\in\mathcal{G}} (1-p)^{\pi(F)} \ \le\ |\mathcal{G}| \cdot e^{-p\min_{F\in\mathcal{G}}|F|/2}. \tag{8.1}$$

So we want few sets, each of them large. The trade-off is between $\log|\mathcal{G}|$ and $p\cdot\min|F|$.

Lemma 8.5. *Let* $t \ge 1$ *and let* $A \subseteq G$ *with* $|A| = t^2$. *Then there is* $A' \subseteq A$ *with* $|A'| = t$ *and* $|A' - A'| \ge t^2/5$.

Proof. Let $n = t^2$ and let A' be a uniformly random t-subset of A. Consider the additive energy $E(A') := \#\{(a_1, a_2, a_3, a_4) \in A'^4 : \ a_1 - a_2 = a_3 - a_4\}$. The *trivial* solutions ($a_1 = a_2, a_3 = a_4$, or $a_1 = a_3, a_2 = a_4$) number at most $2t^2$. A nontrivial solution either has four distinct entries, or satisfies $a_1 = a_4$ or $a_2 = a_3$, since every other coincidence forces a trivial solution. In A^4 there are at most n^3 solutions of the first kind, each lying in A'^4 with probability at most $(t/n)^4$, and at most $2n^2$ of the second kind, each with at least two distinct entries and so lying in A'^4 with probability at most $(t/n)^2$. Hence

$$\mathbb{E}[E(A')] \ \le \ 2t^2 + n^3\Big(\frac{t}{n}\Big)^4 + 2n^2\Big(\frac{t}{n}\Big)^2 = 2t^2 + t^2 + 2t^2 = 5t^2.$$

Fix A' with $E(A') \le 5t^2$. With $r'(x) = \#\{(a_1, a_2) \in A'^2 : a_1 - a_2 = x\}$, the Cauchy–Schwarz inequality gives $t^4 = \big(\sum_x r'(x)\big)^2 \le |A' - A'| \sum_x r'(x)^2 = |A' - A'|\, E(A')$, so $|A' - A'| \ge t^2/5$. □

Proof of Theorem 8.2. Let $t = \lceil 40p^{-1} \log N \rceil$; we may assume $t^2 \le N$. Let

$$\mathcal{G}_t := \big\{(A' - A') \setminus \{0\} : \ A' \subseteq G, \ |A'| = t, \ |A' - A'| \ge t^2/5\big\}.$$

By Lemma 8.5, $\mathcal{G}_t$ covers $\mathcal{H}_{t^2}$. It has $|\mathcal{G}_t| \le \binom{N}{t} \le N^t$ members, each of size at least $t^2/5 - 1$. By (8.1),

$$\mathbb{P}\big(\alpha(G(p)) \ge t^2\big) \le \exp\Big(t \log N - \frac{p\, t^2}{10} + \frac{p}{2}\Big) \le \exp\big(-3t \log N + 1\big) = o(1),$$

since $pt^2 \ge 40\, t \log N$. Hence, with high probability, $\alpha(G(p)) < t^2 = O(p^{-2} \log^2 N)$. □

Cheap witnesses. Independence of a large set A is certified by a much smaller witness: a subset A' of size $t \approx p^{-1} \log N$ that already has $\Omega(t^2)$ distinct differences. There are only N^t witnesses, and each is independent with probability $e^{-\Omega(pt^2)}$. But this cover ignores structure. If $|A - A| \le Kn$ with K small, then $A - A$ is small and the witness $A' - A' \subseteq A - A$ cannot be large. Sets with small doubling need different, more economical witnesses.

8.6 Covering sets of small doubling

Fix n and $K \ge 1$ and consider

$$\mathcal{H}_{n,K} := \big\{A - A : \ |A| = n, \ |A - A| \le Kn\big\}.$$

The heuristic recorded on the board (in red) was: *$\mathcal{H}_{n,K}$ admits a cover by $\exp(\widetilde{O}(K^2))$ sets, each of size $\widetilde{\Omega}(Kn)$.* The precise statement works with several scales ℓ at once.

Theorem 8.6 (Alon–Pham [AP25], version from the lectures). *There is an absolute constant C such that the following holds. Let $n, K \ge 1$ and $L := \lceil \log_2(4K) \rceil$. For every $1 \le \ell \le L$ there is a family $\mathcal{F}_\ell$ of subsets of G such that*

$$|\mathcal{F}_\ell| \le \exp\Big(C \min\big\{2^{2\ell} \log^2 N, \ \sqrt{2^\ell n}\, \log N\big\}\Big) \qquad \text{and} \qquad \min_{F \in \mathcal{F}_\ell} |F| \ \ge \ \frac{2^\ell n}{16\, \ell^2},$$

and for every A with $|A| = n$ and $|A - A| \le Kn$ there are $\ell \le L$ and $F \in \mathcal{F}_\ell$ with $F \subseteq A - A$.

In [AP25, Theorem 6] the second term is $\sqrt{2^\ell n (\log N)^{3/2}}$, which is slightly better, and sumsets $A + A$ are treated as well. The proof has four ingredients: a dyadic pigeonhole argument that finds many *popular differences*; a quick Fourier primer; a *sparse Fourier approximation*, which gives the first bound; and *random subsets*, which give the second.

8.6.1 Popular differences

For $A \subseteq G$ with $|A| = n$, let $r(x) := \#\{(a_1, a_2) \in A^2 : a_1 - a_2 = x\}$. Then $\sum_x r(x) = n^2$, $0 \le r(x) \le n$, and $\operatorname{supp} r = A - A$. For $\ell \ge 1$ define the set of *popular differences*

$$P_\ell := \{x \in G : \ r(x) \ge 2^{-\ell} n\} \ \subseteq \ A - A.$$

Lemma 8.7. *If $|A| = n$ and $|A - A| \le Kn$, then $|P_\ell| \ge \dfrac{2^\ell n}{8\ell^2}$ for some $1 \le \ell \le L = \lceil \log_2(4K) \rceil$.*

Proof. Differences with $r(x) \le n/(4K)$ contribute at most $|A - A| \cdot n/(4K) \le n^2/4$ to $\sum_x r(x)$. Every other x lies in some dyadic class $D_j := \{x : 2^{-j} n < r(x) \le 2^{-j+1} n\}$ with $1 \le j \le L$. Hence $\sum_{j=1}^{L} \sum_{x \in D_j} r(x) \ge \frac{3}{4} n^2$. If we had $\sum_{x \in D_j} r(x) < n^2/(4j^2)$ for all j, the total would be less than $\frac{\pi^2}{24} n^2 < \frac{3}{4} n^2$. So some $j \le L$ has $\sum_{x \in D_j} r(x) \ge n^2/(4j^2)$, and since $r(x) \le 2^{-j+1} n$ on D_j, we get $|P_j| \ge |D_j| \ge 2^j n/(8j^2)$. □

The weights $1/j^2$ make the conclusion independent of K; this is the source of the factor ℓ^2 in Theorem 8.6.

8.6.2 A very quick Fourier primer

Let $\widehat{G}$ be the group of characters $\chi : G \to \mathbb{C}^\times$; for $G = \mathbb{Z}_N$ these are $\chi_\xi(x) = \exp(2\pi i\, \xi x/N)$, $\xi \in \mathbb{Z}_N$, so $\widehat{G} \cong \mathbb{Z}_N$. For $f : G \to \mathbb{C}$ define

$$\widehat{f}(\chi) := \mathbb{E}_{x \in G}\big[f(x)\overline{\chi(x)}\big], \qquad (f * g)(x) := \mathbb{E}_{y \in G}\big[f(y) g(x - y)\big].$$

(This is the convention of [AP25]; many texts in additive combinatorics put $\overline{\chi(x)}$ in the definition of $\widehat{f}$ instead, which replaces $\chi(-x)$ by $\chi(x)$ in the inversion formula below.) Orthogonality of characters gives

$$\text{(inversion)} \quad f(x) = \sum_{\chi} \widehat{f}(\chi)\chi(-x), \qquad \text{(Parseval)} \quad \mathbb{E}_x f(x)\overline{g(x)} = \sum_{\chi} \widehat{f}(\chi)\overline{\widehat{g}(\chi)},$$

$$\text{(convolution)} \quad \widehat{f * g} = \widehat{f}\,\widehat{g}.$$

Identify A with its indicator function and put $\alpha := |A|/N = n/N$. Then $(A * (-A))(x) = \mathbb{E}_y A(y)A(y - x) = r(x)/N$, $\widehat{-A} = \overline{\widehat{A}}$, and so $\widehat{A * (-A)} = |\widehat{A}|^2$, while Parseval gives $\sum_\chi |\widehat{A}(\chi)|^2 = \alpha$. By inversion,

$$\frac{r(x)}{n} = \frac{1}{\alpha}\,(A * (-A))(x) = \sum_{\chi} \frac{|\widehat{A}(\chi)|^2}{\alpha}\,\chi(-x) = \mathbb{E}_{\chi \sim \mu}\big[\chi(-x)\big], \qquad \mu(\chi) := \frac{|\widehat{A}(\chi)|^2}{\alpha}. \tag{8.2}$$

So the normalised representation function r/n is the "characteristic function" of the probability measure μ on $\widehat{G}$.

8.6.3 Sparse Fourier approximation

Formula (8.2) expresses $r(x)/n$ as an average, and averages can be approximated by sampling.

Lemma 8.8. *Let $0 < \eta < 1$ and $m \ge 4\eta^{-2} \log N$, with $N \ge 3$. There are characters $\chi_1, \dots, \chi_m$ such that*

$$\Big|\frac{1}{m}\sum_{i=1}^{m} \operatorname{Re}\chi_i(-x) - \frac{r(x)}{n}\Big| < \eta \qquad \text{for all } x \in G.$$

Proof. Sample $\chi_1, \dots, \chi_m$ independently from μ. For fixed x, the numbers $\operatorname{Re}\chi_i(-x) \in [-1, 1]$ are independent with mean $r(x)/n$ (which is real). By Hoeffding's inequality the deviation is at least η with probability at most $2e^{-m\eta^2/2} \le 2N^{-2}$. A union bound over the N values of x leaves failure probability at most $2/N < 1$. □

Apply this with $\eta = 2^{-\ell-2}$, so $m = \lceil 2^{2\ell+6} \log N \rceil$, and define

$$F(\chi_1, \dots, \chi_m) := \Big\{ x \in G : \ \frac{1}{m} \sum_{i=1}^{m} \operatorname{Re} \chi_i(-x) > 2^{-\ell-1} \Big\}.$$

If $x \in P_\ell$ then $r(x)/n \geq 2^{-\ell}$, so the average exceeds $2^{-\ell} - 2^{-\ell-2} > 2^{-\ell-1}$ and $x \in F$. If $x \notin A - A$ then $r(x) = 0$, so the average is below $2^{-\ell-2}$ and $x \notin F$. Therefore

$$P_\ell \ \subseteq \ F(\chi_1, \dots, \chi_m) \ \subseteq \ A - A,$$

and F is determined by m characters, giving at most $N^m \leq \exp(2^{2\ell+7} \log^2 N)$ possibilities (Figure 8.2). Sets A whose differences are concentrated on few popular values (small ℓ) are thus certified by a short Fourier description.

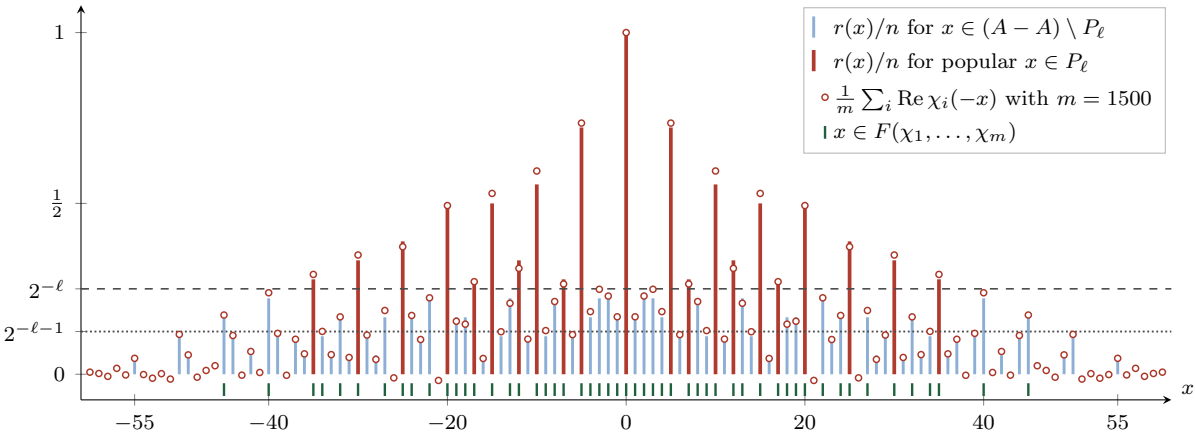


Figure 8.2: A real example in $\mathbb{Z}_{211}$: A is an arithmetic progression of length 10 with difference 5, together with 8 random elements of $[0, 60)$, so $n = 18$ and $|A - A| = 87$ (the stems; $\operatorname{supp} r = A - A \subseteq [-55, 55]$). Here $\ell = 2$: stems reaching the dashed line $2^{-\ell} = \frac{1}{4}$ are the popular differences. The open circles show the empirical average over $m = 1500$ characters sampled from μ; at every x it is within $2^{-\ell-2}$ of $r(x)/n$. The green ticks mark the set F where the average exceeds the dotted line $2^{-\ell-1}$, so that $P_\ell \subseteq F \subseteq A - A$.

8.6.4 Random subsets

For large ℓ the Fourier description is too long, and the cheaper witness is a random subset, as in Lemma 8.5.

Lemma 8.9. *For every A with $|A| = n$ and every $\ell \geq 1$ there is $A' \subseteq A$ with $|A'| \leq 7\sqrt{2^\ell n}$ and $|(A' - A') \cap P_\ell| \geq |P_\ell|/2$.*

Proof. Let $\theta := \min\{1, \sqrt{3 \cdot 2^\ell / n}\}$ and keep each element of A independently with probability θ, forming A'. (If $\theta = 1$ then $A' = A$ and $|A| = n \leq \sqrt{3 \cdot 2^\ell n}$.) Let $x \in P_\ell$, $x \neq 0$, and let Γ_x be the graph on A with edges $\{a, a + x\}$ $(a, a + x \in A)$. If $2x \neq 0$, then Γ_x has $r(x)$ edges and maximum degree at most 2, so it is a union of paths and cycles and contains a matching of size at least $r(x)/3$. If $2x = 0$, then Γ_x is itself a matching with $r(x)/2$ edges. If both ends of one matching edge survive, then $x \in A' - A'$. Hence

$$\mathbb{P}(x \notin A' - A') \leq (1 - \theta^2)^{r(x)/3} \leq \exp\big(-\theta^2 \, 2^{-\ell} n/3\big) \leq e^{-1},$$

and also $\mathbb{P}(0 \notin A' - A') = \mathbb{P}(A' = \emptyset) \leq e^{-\theta n} \leq e^{-1}$. Thus $\mathbb{E}|P_\ell \setminus (A' - A')| \leq |P_\ell|/e$, so by Markov's inequality $\mathbb{P}\big(|P_\ell \setminus (A' - A')| > |P_\ell|/2\big) \leq 2/e < 0.74$, while $\mathbb{P}(|A'| > 4\theta n) \leq \frac{1}{4}$. With positive probability neither happens, and then $|A'| \leq 4\sqrt{3 \cdot 2^\ell n} \leq 7\sqrt{2^\ell n}$. □

Proof of Theorem 8.6. For $1 \le \ell \le L$ let $m_\ell := \lceil 2^{2\ell+6} \log N \rceil$ and set

$$\mathcal{F}_\ell^{(1)} := \{F(\chi_1, \dots, \chi_{m_\ell}) : \ \chi_i \in \widehat{G}\}, \qquad |\mathcal{F}_\ell^{(1)}| \le N^{m_\ell} \le e^{2^{2\ell+7} \log^2 N},$$
$$\mathcal{F}_\ell^{(2)} := \{A' - A' : \ A' \subseteq G, \ |A'| \le 7\sqrt{2^\ell n}\}, \qquad |\mathcal{F}_\ell^{(2)}| \le (N+1)^{7\sqrt{2^\ell n}} \le e^{14\sqrt{2^\ell n} \log N},$$

discarding in both families the sets of size less than $2^\ell n/(16\ell^2)$. Let $\mathcal{F}_\ell$ be the smaller of the two families. Given A with $|A| = n$ and $|A - A| \le Kn$, Lemma 8.7 provides $\ell \le L$ with $|P_\ell| \ge 2^\ell n/(8\ell^2)$. The sparse Fourier approximation gives $F \in \mathcal{F}_\ell^{(1)}$ with $P_\ell \subseteq F \subseteq A - A$, and Lemma 8.9 gives $F = A' - A' \in \mathcal{F}_\ell^{(2)}$ with $F \subseteq A - A$ and $|F| \ge |P_\ell|/2$. Both have size at least $2^\ell n/(16\ell^2)$, so whichever family is $\mathcal{F}_\ell$ contains a subset of $A - A$. □

8.7 Proof of the $p^{-3/2}$ bound

Proof of Theorem 8.4. Let $n := \lceil K_0\, p^{-3/2} \log^6 N \rceil$ for a large constant K_0; we may assume $n \le N$. Every n-set satisfies $|A - A| \le n^2$, so we apply Theorem 8.6 with $K = n$. Then $L = \lceil \log_2(4n) \rceil \le 2 \log N$ for large N, and the families $\mathcal{F}_\ell$ together cover $\mathcal{H}_n$. By (8.1), with C from Theorem 8.6,

$$\mathbb{P}\big(\alpha(G(p)) \ge n\big) \le \sum_{\ell=1}^{L} \exp(E_\ell), \qquad E_\ell := C \min\big\{2^{2\ell} \log^2 N, \ \sqrt{2^\ell n} \log N\big\} - \frac{p\, 2^\ell n}{32\ell^2} + \frac{1}{2}.$$

We claim that the first term of E_ℓ is at most $p\, 2^\ell n/(64\ell^2)$ for every $\ell \le L$. We use $\ell \le 2 \log N$.

- If $2^\ell \le p^{-1/2}$, use the Fourier bound: $C 2^{2\ell} \log^2 N \le p 2^\ell n/(64\ell^2)$ holds as soon as $n \ge 64C\, \ell^2\, 2^\ell\, p^{-1} \log^2 N$, which follows from $n \ge 256\, C\, p^{-3/2} \log^4 N$.
- If $2^\ell > p^{-1/2}$, use the random subset bound: $C\sqrt{2^\ell n} \log N \le p 2^\ell n/(64\ell^2)$ is equivalent to $2^\ell n \ge (64C)^2 \ell^4 p^{-2} \log^2 N$, which follows from $n \ge 16(64C)^2 p^{-3/2} \log^6 N$.

Hence, using $2^\ell/\ell^2 \ge \frac{8}{9}$ for all $\ell \ge 1$,

$$E_\ell \ \le \ -\frac{p\, 2^\ell n}{64\ell^2} + \frac{1}{2} \ \le \ -\frac{pn}{72} + \frac{1}{2}.$$

Since $pn \ge K_0 p^{-1/2} \log^6 N \ge K_0 \log^6 N$, the sum of the at most $2 \log N$ terms is $o(1)$. So, with high probability, $\alpha(G(p)) < n = O(p^{-3/2} \log^6 N)$. □

The exponent $\frac{3}{2}$ comes from balancing the two constructions at $2^\ell \approx p^{-1/2}$. Very structured sets (small ℓ, few but very popular differences) have a short Fourier description. Unstructured sets (large ℓ) are certified by a random sample. The hardest sets are in between. The power of $\log N$ was not optimised; [AP25] obtain $(\log N)^{19/4}$.

Outlook. The same computation shows that a cover of $\mathcal{H}_{n,K}$ at every scale ℓ by families $\mathcal{F}_\ell$ with $\log |\mathcal{F}_\ell| = \widetilde{O}(2^\ell)$ and $\min_{F \in \mathcal{F}_\ell} |F| = \widetilde{\Omega}(2^\ell n)$ would give $\alpha(G(p)) = \widetilde{O}(p^{-1})$, which is Alon's conjecture. This is formulated in [AP25, Conjecture 15]. It is a purely additive-combinatorial statement: how efficiently can one certify that a set has a large difference set? In the spirit of the Kahn–Kalai theorem, the truth is decided by first moment obstructions, and the task is to construct the right ones.

8.8 Notes and references

Random Cayley graphs and the bound of Theorem 8.2 are due to Alon [Alo07, Alo13], who also conjectured the $\widetilde{O}(p^{-1})$ bound. Counting sets with small sumset, and the clique number of

random Cayley graphs for constant p, go back to Green [Gre05]; the $O(\log N \log\log N)$ bound for every abelian group is due to Conlon, Fox, Pham and Yepremyan [CFPY24]. Theorem 8.4 and the covering theorem are from Alon and Pham [AP25]: Lemma 8.7 is their Lemma 10, the sparse Fourier approximation is the argument of their Lemma 9 and Theorems 11–12, and the general statement, which also covers sumsets $A + A$ and sharper logarithmic factors, is their Theorem 6. Theorem 8.6 is the simplified version presented in the lectures, and Lemma 8.5 is a standard additive energy argument.

A Probabilistic and Analytic Tools

This appendix collects the elementary inequalities used throughout the notes.

A.1 Elementary inequalities

- *Markov's inequality.* If $X \geq 0$ and $a > 0$, then $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. For X taking values in $\{0, 1, 2, \dots\}$ this gives $\mathbb{P}(X \neq 0) \leq \mathbb{E}X$ (Lemma 1.1).
- *Chebyshev's inequality.* $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \operatorname{Var} X/a^2$; in particular $\mathbb{P}(X = 0) \leq \operatorname{Var} X/(\mathbb{E}X)^2 = \mathbb{E}[X^2]/(\mathbb{E}X)^2 - 1$.
- *Union bound.* $\mathbb{P}(\bigcup_i E_i) \leq \sum_i \mathbb{P}(E_i)$.
- *Exponential bounds.* $1 - u \leq e^{-u}$ for all real u, and $1 - u \geq e^{-2u}$ for $u \in [0, \frac{1}{2}]$. Hence $(1-p)^m \leq e^{-pm}$, and $(1-p)^m \geq e^{-2pm}$ for $p \leq \frac{1}{2}$.
- *Binomial coefficients.* $(n/k)^k \leq \binom{n}{k} \leq (en/k)^k$ for $1 \leq k \leq n$, and $\sum_{j \leq J} \binom{N}{j} \leq (N+1)^J$.
- *Factorials.* $n! \geq (n/e)^n$. Consequently, for $0 \leq s \leq n$, the product of the s largest factors of $n!$ satisfies $n!/(n-s)! \geq (n!)^{s/n} \geq (n/e)^s$.
- *Cauchy–Schwarz.* $\big(\sum_x a_x b_x\big)^2 \leq \sum_x a_x^2 \sum_x b_x^2$; with $b_x = \mathbf{1}\{a_x \neq 0\}$ this gives $\big(\sum_x a_x\big)^2 \leq |\operatorname{supp} a| \sum_x a_x^2$.

A.2 Concentration

Lemma A.1 (Hoeffding's inequality [Hoe63]). *Let $V_1, \dots, V_m$ be independent random variables with values in $[a, b]$, and let $S = \sum_i V_i$. For every $t \geq 0$,*

$$\mathbb{P}\big(|S - \mathbb{E}S| \geq t\big) \;\leq\; 2\exp\Big(-\frac{2t^2}{m(b-a)^2}\Big).$$

In particular, if $V_i \in [-1, 1]$ and $\eta > 0$, then $\mathbb{P}\big(|\frac{1}{m}S - \frac{1}{m}\mathbb{E}S| \geq \eta\big) \leq 2e^{-m\eta^2/2}$.

This is the form used in the sparse Fourier approximation (Lemma 8.8).

A.3 Linear programming duality

Theorem A.2 (Strong duality). *Let A be a real matrix and b, c real vectors. If the linear program*

$$\min\{c^\top x : \; Ax \geq b, \; x \geq 0\}$$

has a feasible solution and a finite optimum, then so does its dual $\max\{b^\top y : \; A^\top y \leq c, \; y \geq 0\}$, and the two optima are equal.

In Chapter 4 the variables x are the weights $g(G)$ of a fractional cover, the constraints $Ax \geq b$ say that every $H \in \mathcal{H}$ is covered, and $c_G = p^{|G|}$; the dual variables y are the weights $\lambda(H)$, which after normalisation become a spread measure. See, for instance, Schrijver [Sch86] for a proof.

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